Chapter 9

GPIO Matrix and IO MUX

9.1 Overview

GPIO matrix is a hardware structure that allows dynamic remapping between the GPIO pins and peripheral input/output signals to provide greater flexibility. IO MUX (Input/output multiplexing) is a technology that allows the same pin to perform different functions at different times, allowing dynamic switching of pin functions through configuration. ESP32-P4 provides two groups of IO MUX and GPIO matrix:

- HP GPIO matrix and HP IO MUX
- LP GPIO matrix and LP IO MUX

The ESP32-P4 chip features 55 GPIO pins, including 16 low-power (LP) GPIO pins and 39 high-performance (HP) GPIO pins. Each pin can be used as a general-purpose I/O, or be connected to an internal peripheral signal.

- Through HP GPIO matrix and HP IO MUX, HP peripheral input signals can be from any GPIO pins, and HP peripheral output signals can be routed to any GPIO pins.
- Through LP GPIO matrix and LP IO MUX, LP peripheral input signals can be from any LP GPIO pins, and LP peripheral output signals can be routed to any LP GPIO pins.

Together these modules provide highly configurable I/O. The 55 GPIO pins are numbered from GPIO0 to GPIO54.

- LP GPIO pins (GPIO0 ~ GPIO15) can be used by either HP or LP peripherals.
- HP GPIO pins (GPIO16 ~ GPIO54) can be used only by HP peripherals.

The ESP32-P4 chip has dedicated pins for external flash and in-package PSRAM. Such pins can not be used for other purpose. See *ESP32-P4 Datasheet* > Section *Pin Mapping Between Chip and Flash/PSRAM*.

Unless otherwise specified, GPIO matrix refers to both LP GPIO matrix and HP GPIO matrix, so as to IO MUX.

9.2 Features

9.2.1 HP GPIO Matrix and HP IO MUX

HP GPIO matrix has the following features:

- A full-switching matrix between HP peripheral input/output signals and the GPIO pins
- 222 HP peripheral input signals sourced from the input of any GPIO pins
- 232 HP peripheral output signals routed to the output of any GPIO pins

- Signal synchronization for HP peripheral inputs based on **HP IO MUX operating clock**
- GPIO Filter hardware for input signal filtering
- Glitch Filter hardware for second-time filtering on input signal
- Sigma delta modulated (SDM) output
- GPIO simple input and output
- HP GPIO Wakeup

HP IO MUX has the following features:

- Control of 55 GPIOs (GPIO0 ~ GPIO54) for HP peripherals.
- A configuration register IO_MUX_GPIO*n*_REG provided for each GPIO pin, to control the pin's input/output, pull-up/pull-down, drive strength, and function selection.
- Better high-frequency digital performance achieved by routing some digital signals (SPI, EMAC) directly from HP IO MUX to peripherals.

9.2.2 LP GPIO Matrix and LP IO MUX

LP GPIO matrix has the following features:

- A full-switching matrix between the LP peripheral input/output signals and the LP GPIO pins
- 14 LP peripheral input signals sourced from the input of any LP GPIO pins
- 14 LP peripheral output signals routed to the output of any LP GPIO pins
- GPIO Filter hardware for input signal filtering
- GPIO simple input and output
- LP GPIO Wakeup

LP IO MUX has the following feature:

- Control of 16 LP GPIO pins (GPIO0 ~ GPIO15) for LP peripherals.
- A configuration register LP_IOMUX_PAD*n*_REG provided for each LP GPIO pin, to control the pin's input/output, pull-up/pull-down, drive strength, function selection, and IO MUX selection.

9.3 Architectural Overview

Figure 9.3-1 shows in details how HP GPIO matrix, HP IO MUX, LP GPIO matrix, and LP IO MUX route signals from pins to peripherals, and from peripherals to pins.

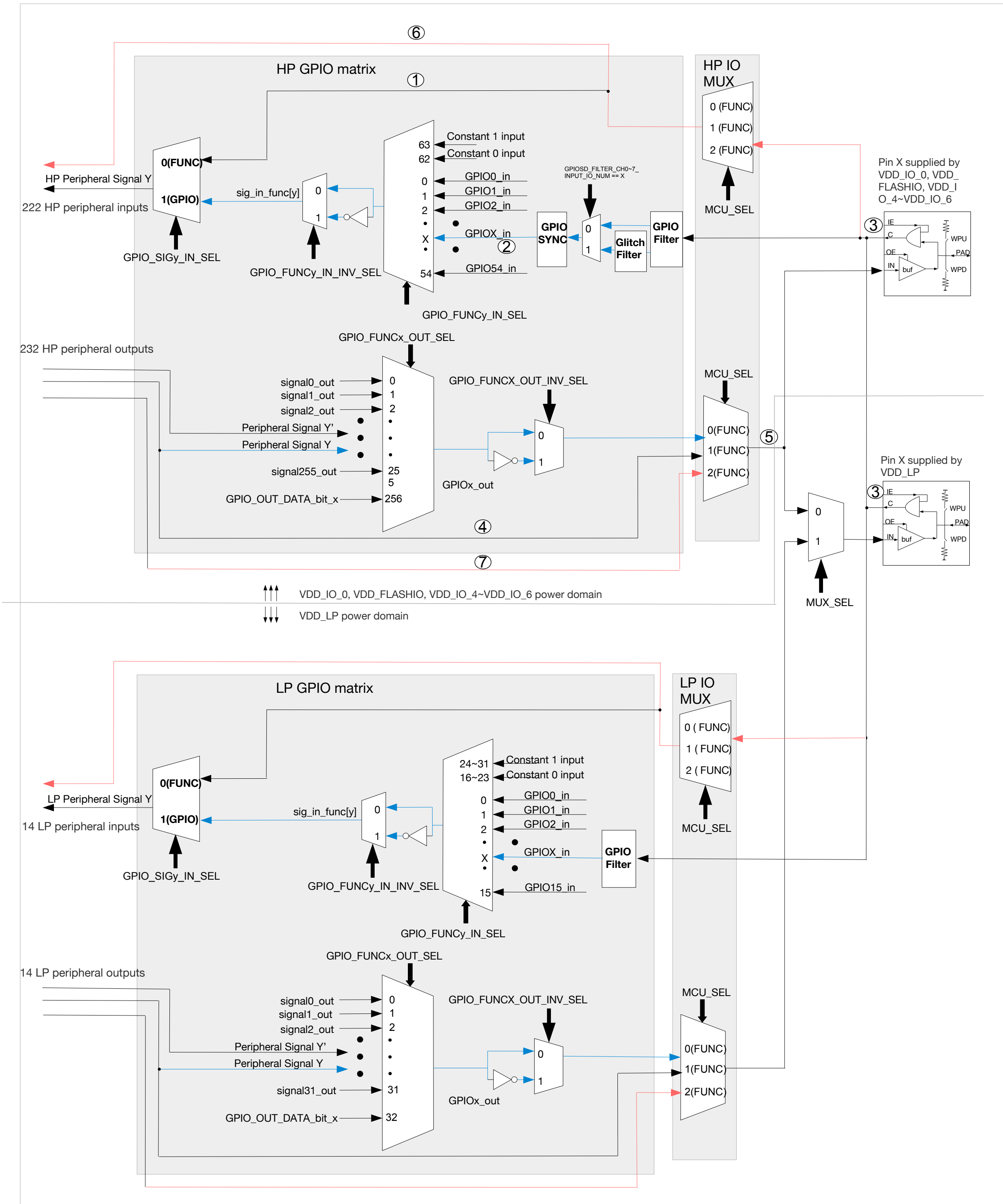

Figure 9.3-1. Architecture of HP GPIO Matrix, HP IO MUX, LP GPIO Matrix, and LP IO MUX

The following points explain the areas marked numerically in the figure above, taking the HP system as an example:

- ① Part of peripheral input signals (marked "yes" in column "Direct input through HP IO MUX" in Table

9.12-1) can be routed to the peripherals via HP IO MUX, or via HP GPIO matrix, while the other input signals can only be routed to the peripherals via HP GPIO matrix.

- ② There are only 55 inputs from GPIO SYNC to HP GPIO matrix, since ESP32-P4 provides 55 GPIO pins in total.
- ③ The pins supplied by VDD_IO_0, VDD_FLASHIO, VDD_IO_4 ~ VDD_IO_6, and VDD_LP are controlled by the signals: IE, OE, HYS, WPU, and WPD.
- ④ Part of peripheral outputs (marked "yes" in column "Direct output through HP IO MUX" in Table 9.12-1) can be routed to pins via HP IO MUX or via HP GPIO matrix, while the other output signals can only be routed to pins via HP GPIO matrix.
- ⑤ There are 55 outputs (corresponding to GPIO pin *X*: 0 ~ 54) from HP GPIO matrix to HP IO MUX.
- ⑥ There are peripheral inputs only through HP IO MUX.
- ⑦ There are peripheral outputs only through HP IO MUX.

Figure 9.3-2 shows the internal structure of a pad, which is an electrical interface between the chip logic and the GPIO pin. The structure is applicable to all 55 GPIO pins and can be controlled using IE, OE, WPU, and WPD signals. For the configuration of these signals, see IO_MUX_GPIO*x*_REG orLP_IOMUX_PAD*x*_REG.

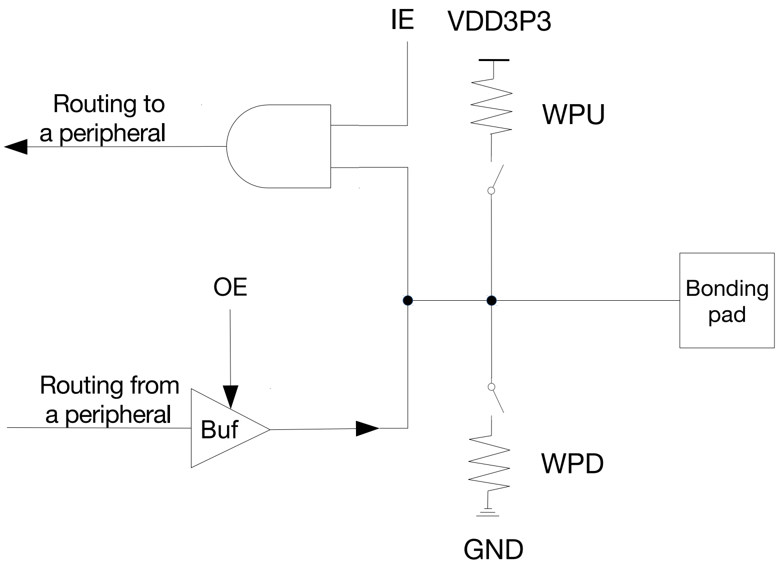

Figure 9.3-2. Internal Structure of a Pad

- IE: input enable
- OE: output enable
- WPU: internal weak pull-up resistor
- WPD: internal weak pull-down resistor
- Bonding pad: a terminal point of the chip logic used to make a physical connection from the chip die to GPIO pin in the chip package

9.4 Peripheral Input via GPIO Matrix

9.4.1 Overview

To receive a peripheral input signal via HP GPIO matrix,

- configure the matrix to source the peripheral input signal from one of the 55 GPIOs (0 ~ 54), see Table 9.12-1.
- configure the peripheral signal to receive input signal via HP GPIO matrix.
- configure the GPIO pin to be controlled by HP IO MUX.

For detailed configuration, see Figure 9.3-1 and Section 9.4.7.

To receive a peripheral input signal via LP GPIO matrix,

- configure the matrix to source the peripheral input signal from one of the 16 GPIOs (0 ~ 15), see Table 9.13-1.
- configure the peripheral signal to receive input signal via LP GPIO matrix.
- configure the LP GPIO pin to be controlled by LP IO MUX.

For detailed configuration, see Figure 9.3-1 and Section 9.4.7.

As shown in Figure 9.3-1, when GPIO matrix is used to input a signal from the pin, all external input signals are sourced from the GPIO pins and then filtered by the GPIO Filter, as shown in Step 2 in Section 9.4.7.

The Glitch Filter is only available in HP GPIO matrix. The Glitch Filter hardware can filter eight of the output signals from the GPIO Filter, and the other unselected signals go directly to the GPIO SYNC hardware, as shown in Step 3 in Section 9.4.7.

All signals filtered by the GPIO Filter hardware or the Glitch Filter hardware are synchronized by the GPIO SYNC hardware to IO MUX operating clock and then enter the GPIO matrix, see Section 9.4.2. Such signal filtering and synchronization features apply to all GPIO matrix signals but do not apply when using the IO MUX.

9.4.2 Signal Synchronization

Only HP GPIO matrix supports this signal synchronization function. Figure 9.4-1 shows the functionality of GPIO SYNC. In the figure, negative sync and positive sync mean GPIO input is synchronized on falling edge and on rising edge of HP IO MUX operating clock respectively.

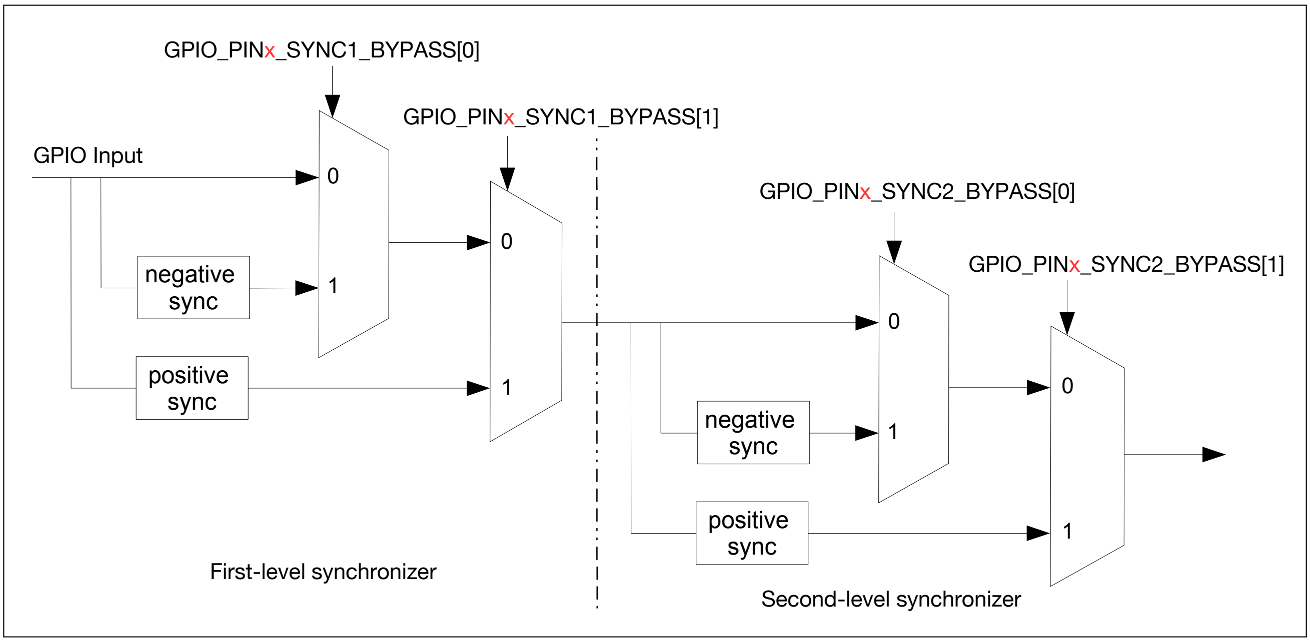

Figure 9.4-1. GPIO Input Synchronized on Rising Edge or on Falling Edge of HP IO MUX Operating Clock

The synchronization function is disabled by default by the synchronizer. But when an asynchronous peripheral signal is connected to the pin, the signal should be synchronized by the two-level synchronizer (i.e., the first-level synchronizer and the second-level synchronizer as shown in Figure 9.4-1) to lower the probability of causing metastability.

9.4.3 GPIO Filter

Both the HP GPIO matrix and LP GPIO matrix support GPIO Filter function. When this function is enabled, only signals with a valid width of more than two clock cycles can be sampled, as illustrated in Figure 9.4-2.

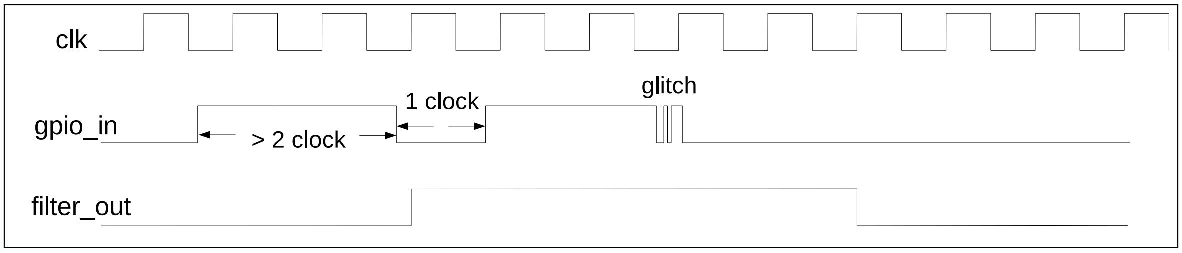

Figure 9.4-2. GPIO Filter Timing of GPIO Input Signals

9.4.4 Glitch Filter

The Glitch Filter function is exclusive to the HP GPIO matrix. The Glitch Filter hardware supports eight channels, each of which selects one signal from the 55 (0 ~ 54) output signals generated by the GPIO Filter hardware. It then conducts a second round of filtering on the selected signal. This Glitch Filter hardware can be used to filter slow-speed signals. For more information, see Step 3 in Section 9.4.7.1.

9.4.5 Simple GPIO Input

Both the HP GPIO matrix and LP GPIO matrix support the Simple GPIO Input function. Enabling this function allows the direct reading of the input value of a GPIO pin at any time, without the need to route the GPIO input to any peripherals.

For HP GPIO matrix, to implement simple GPIO input, follow the steps below:

- Set IO_MUX_GPIO*x*_MCU_IE in IO_MUX_GPIO*x*_REG, to enable pin input.
- Read the GPIO input from GPIO_IN_REG[*x*].

For LP GPIO matrix, to implement simple GPIO input, follow the steps below:

- Set LP_IOMUX_PAD*x*_FUN_IE in LP_IO_MUX_PAD*x*_REG, to enable pin input.
- Read the GPIO input from LP_GPIO_IN_REG[*x*].

9.4.6 GPIO Wakeup

9.4.6.1 HP GPIO Wakeup

The HP GPIO wakeup feature is designed to awaken the system from Light-sleep when the HP system is in clock gating mode. HP GPIO wakeup is not available when the HP system is powered down.

GPIO0 ~ GPIO54 can be used to generate HP GPIO wakeup by enabling GPIO_PIN*x*_WAKEUP_ENABLE (*x*: 0 ~ 54).

HP GPIO wakeup supports both high-voltage sensitive and low-voltage sensitive wakeups, with specific conditions based on the GPIO configuration.

Table 9.4-1. HP GPIO Wakeup Signal Trigger and Clear Conditions

GPIO_PIN*x*_INT_TYPE[1,2]	Wakeup is generated when	Wakeup is cleared when
5	the input GPIO is high-level.	the input GPIO is low-level.
4	the input GPIO is low-level.	the input GPIO is high-level.

[1] *x* ranges from 0 to 54.
[2] If the register is configured to any other value, HP GPIO wakeup feature is disabled.

9.4.6.2 LP GPIO Wakeup

The LP GPIO wakeup feature is designed to awaken the system from Deep-sleep when the LP peripherals are not in power gating mode. LP GPIO wakeup is not available when the LP peripherals are powered down.

GPIO0 ~ GPIO15 can be used to generate LP GPIO wakeup by enabling LP_GPIO_PIN*n*_WAKEUP_ENABLE (*n*: 0 ~ 15).

LP GPIO wakeup supports posedge sensitive, negedge sensitive, both posedge and negedge sensitive, high-voltage sensitive and low-voltage sensitive wakeups, with specific conditions based on the GPIO configuration.

Table 9.4-2. LP GPIO Wakeup Signal Trigger and Clear Conditions

LP_GPIO_PINx_INT_TYPE[1,2]	Wakeup is generated when	Wakeup is cleared when
1	the input GPIO toggles from low-level to high-level.	LP_GPIO_PINx_EDGE_WAKEUP_CLR is set.
2	the input GPIO toggles from high-level to low-level.	LP_GPIO_PINx_EDGE_WAKEUP_CLR is set.
3	the input GPIO toggles from high-level to low-level, or vice versa.	LP_GPIO_PINx_EDGE_WAKEUP_CLR is set.
4	the input GPIO is low-level.	the input GPIO is high-level.
5	the input GPIO is high-level.	the input GPIO is low-level.

[1] *x* ranges from 0 to 15.

[2] If the register is configured to any other value, LP GPIO wakeup feature is disabled.

9.4.7 Programming Procedure

9.4.7.1 HP GPIO Matrix

To read GPIO pin *X*[1] into HP peripheral signal *Y*, follow the steps below:

1. Configure GPIO_FUNC*y*_IN_SEL_CFG_REG corresponding to HP peripheral signal *Y* in HP GPIO matrix:
   - Set GPIO_SIG*y*_IN_SEL to enable peripheral signal input via HP GPIO matrix.
   - Set GPIO_FUNC*y*_IN_SEL to the desired GPIO pin, i.e., *X* here.

   Note that some peripheral signals have no valid GPIO_SIG*y*_IN_SEL bit, namely, these peripherals can only receive input signals via HP GPIO matrix.

2. Optionally enable the GPIO Filter for pin input signals by setting IO_MUX_GPIO*n*_FILTER_EN.

3. Enable Glitch Filter feature as follows:
   - Configure GPIO_EXT_FILTER_CH*n*_INPUT_IO_NUM to *m*. *n* (0 ~ 7) represents the channel number. *m* (0 ~ 54) represents the GPIO pin number.
   - Configure GPIO_EXT_FILTER_CH*n*_WINDOW_WIDTH to **VALUE1** and GPIO_EXT_FILTER_CH*n*_WINDOW_THRES to **VALUE2**. During **VALUE1** + 1 cycles, if there are **VALUE2** + 1 input signals that do not match the current output signal value, the Glitch Filter hardware inverts the output signal. GPIO_EXT_FILTER_CH*n*_WINDOW_WIDTH and GPIO_EXT_FILTER_CH*n*_WINDOW_THRES can be configured to the same value **VALUE3**, then only signals with a width greater than **VALUE3** + 1 clock cycles will be sampled.
   - Set GPIO_EXT_FILTER_CH*n*_EN to enable channel *n*.

   An example is shown in Figure 9.4-3, where GPIO_EXT_FILTER_CH*x*_WINDOW_WIDTH is configured to 3 and GPIO_EXT_FILTER_CH*x*_WINDOW_THRES to 2. The output signal value (signal_out) keeps as "0" in the four clock cycles before T1. The input signal value (signal_in) has been "1" for three clock cycles in the same period, then the output signal is inverted to "1" after T1.

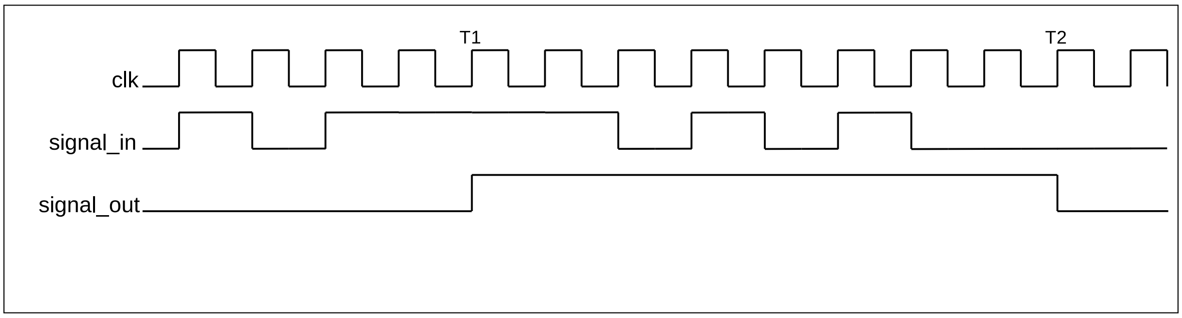

Figure 9.4-3. Glitch Filter Timing Example

4. Synchronize GPIO input signals. To do so, please set GPIO_PIN*x*_REG corresponding to GPIO pin *X* as follows:
   - Set GPIO_PIN*x*_SYNC1_BYPASS to enable input signal synchronized on rising edge or on falling edge in the first-level synchronization, see Figure 9.4-1.
   - Set GPIO_PIN*x*_SYNC2_BYPASS to enable input signal synchronized on rising edge or on falling edge in the second-level synchronization, see Figure 9.4-1.
5. Configure HP IO MUX register to enable pin input. For this end, please set IO_MUX_GPIO*x*_REG corresponding to GPIO pin *X* as follows:
   - Set IO_MUX_GPIO*x*_FUN_IE to enable input[2].
   - Set or clear IO_MUX_GPIO*x*_FUN_WPU and IO_MUX_GPIO*x*_FUN_WPD as desired to enable or disable pull-up and pull-down resistors.

For example, to connect UART0 TXD input signal[3] (uart0_rxd_pad_in, signal index 10) to GPIO7, please follow the steps below.

1. Set GPIO_SIG10_IN_SEL in GPIO_FUNC10_IN_SEL_CFG_REG to enable peripheral signal input via HP GPIO matrix.
2. Set GPIO_FUNC10_IN_SEL in GPIO_FUNC10_IN_SEL_CFG_REG to 7, i.e., select GPIO7.
3. Set IO_MUX_GPIO7_FUN_IE in IO_MUX_GPIO7_REG to enable pin input.

Note:

1. One input pin can be connected to multiple peripheral input signals.
2. The input signal can be inverted by configuring GPIO_FUNC*y*_IN_INV_SEL.
3. It is possible to have a HP peripheral read a constantly low or constantly high input value without connecting this input to a pin. This can be done by selecting a special GPIO_FUNC*y*_IN_SEL input, instead of a GPIO number:
   - When GPIO_FUNC*y*_IN_SEL is set to 62, input signal is always 0.
   - When GPIO_FUNC*y*_IN_SEL is set to 63, input signal is always 1.

9.4.7.2 LP GPIO Matrix

The programming procedure for peripheral input via the LP GPIO matrix resembles that of Section 9.4.7.1. However, it's worth noting that:

- the LP GPIO matrix lacks a Glitch Filter function.
- The control and status registers must use LP GPIO related registers. See Section 9.19.4 and Section 9.19.5.

Note:
It is possible to have a LP peripheral read a constantly low or constantly high input value without connecting this input to a pin. This can be done by selecting a special LP_GPIO_SIG*y*_IN_SEL input, instead of a GPIO number:

- When LP_GPIO_SIG*y*_IN_SEL is set to 16, input signal is always 0.
- When LP_GPIO_SIG*y*_IN_SEL is set to 24, input signal is always 1.

9.5 Peripheral Output via GPIO Matrix

9.5.1 Overview

To output a signal from a peripheral via GPIO matrix,

- configure the HP GPIO matrix to route HP peripheral output signals (only signals with a name assigned in the column "Output signal" in Table 9.12-1) to one of the 55 GPIOs (0 ~ 54).
- Or configure the LP GPIO matrix to route LP peripherals (only signals with a name assigned in the column "Output signal" in Table 9.13-1) to one of the 16 LP GPIOs (0 ~ 15).
- The output signal is routed from the peripheral into GPIO matrix and then into IO MUX. IO MUX must be configured to set the chosen pin to GPIO function. This enables the GPIO output signal to be connected to the pin.

Note:
There is a range of peripheral output signals (250 ~ 255 in Table 9.12-1) which are not connected to any peripheral, but to the input signals (250 ~ 255) directly.

9.5.2 Simple GPIO Output

GPIO matrix can also be used for simple GPIO output. For this case, one GPIO pin can be configured to directly output the desired value, without routing any peripheral output to this pin.

Follow the steps below to configure HP GPIO matrix for simple GPIO output:

- Set GPIO_FUNC*x*_OUT_SEL with a special peripheral index 256 (0x100).
- Configure the corresponding bit in GPIO_OUT_REG/GPIO_OUT1_REG to the desired GPIO output value.

Note:

- GPIO_OUT_REG[0] ~ GPIO_OUT_REG[31] correspond to GPIO0 ~ GPIO31, and GPIO_OUT1_REG[0] ~ GPIO_OUT1_REG[22] to GPIO32 ~ GPIO54.
- Recommended operation: use GPIO_OUT/OUT1_W1TS and GPIO_OUT/OUT1_W1TC to set or clear GPIO_OUT/OUT1_REG.

Follow the steps below to configure LP GPIO matrix for simple GPIO output:

- Set LP_GPIO_FUNCx_OUT_SEL with a special peripheral index 32 (0x20).
- Configure the corresponding bit in LP_GPIO_OUT_REG to the desired GPIO output value.

> **Note:**
> Recommended operation: use LP_GPIO_OUT_DATA_W1TS and LP_GPIO_OUT_DATA_W1TC to set or clear LP_GPIO_OUT_REG.

9.5.3 Sigma Delta Modulated Output (SDM)

9.5.3.1 Functional Description

Eight HP peripheral output signals (index: 72 ~ 79 in Table 9.12-1) support 1-bit second-order sigma delta modulation. By default the output is enabled for these eight channels. This Sigma Delta modulator can also output PDM (pulse density modulation) signal with configurable duty cycle. The function is:

$$H(z) = \mathrm{X(z)z}^{-1} + \mathrm{E(z)(1\text{-}z}^{-1})^2$$

E(z) is quantization error and X(z) is the input.
This modulator supports scaling down of IO MUX operating clock by divider 1 ~ 256:

- Set GPIO_EXT_SD_FUNCTION_CLK_EN to enable the modulator clock.
- Configure GPIO_EXT_SDn_PRESCALE (*n* = 0 ~ 7 for the eight channels).

After scaling, the clock cycle is equal to one pulse output cycle from the modulator.

GPIO_EXT_SDn_IN is a signed number with a range of [-128, 127] and is used to control the duty cycle[1] of PDM output signal.

- GPIO_EXT_SDn_IN = -128, the duty cycle of the output signal is 0%.
- GPIO_EXT_SDn_IN = 0, the duty cycle of the output signal is near 50%.
- GPIO_EXT_SDn_IN = 127, the duty cycle of the output signal is near 100%.

The formula for calculating PDM signal duty cycle is shown as below:

$$Duty\_Cycle = \frac{GPIO\_EXT\_SDn\_IN + 128}{256}$$

> **Note:**
> For PDM signals, duty cycle refers to the percentage of high level cycles to the whole statistical period (several pulse cycles, for example, 256 pulse cycles).

9.5.3.2 SDM Configuration

The configuration of SDM is shown below:

- Route one of SDM outputs to a pin via HP GPIO matrix, see Section 9.5.4.1.

- Enable the modulator clock by setting GPIO_EXT_SD_FUNCTION_CLK_EN.
- Configure the divider value by setting GPIO_EXT_SDn_PRESCALE.
- Configure the duty cycle of SDM output signal by setting GPIO_EXT_SDn_IN.

9.5.4 Programming Procedure

9.5.4.1 HP GPIO Matrix

The output signals with a name assigned in the column "Output signal" in Table 9.12-1 can be set to go through the HP GPIO matrix into HP IO MUX and then to a pin. Figure 9.3-1 illustrates the configuration.

To output peripheral signal Y to a particular GPIO pin X, follow the steps below:

1. Configure GPIO_FUNCx_OUT_SEL_CFG_REG and GPIO_ENABLE_REG[x] corresponding to GPIO pin X in HP GPIO matrix. Recommended operation: use corresponding W1TS (write 1 to set) and W1TC (write 1 to clear) registers to set or clear GPIO_ENABLE_REG.
   - Set the GPIO_FUNCx_OUT_SEL field in register GPIO_FUNCx_OUT_SEL_CFG_REG to the index of the desired peripheral output signal Y.
   - If the signal should always be enabled as an output, set the GPIO_FUNCx_OE_SEL bit in register GPIO_FUNCx_OUT_SEL_CFG_REG and the bit in register GPIO_ENABLE_W1TS_REG, corresponding to GPIO pin X. To have the output enable signal decided by internal logic (for example, the spi2_dqs_pad_oe in column "Output enable signal when GPIO_FUNCn_OE_SEL = 0" in Table 9.12-1), clear the GPIO_FUNCx_OE_SEL bit instead.
   - Set the corresponding bit in register GPIO_ENABLE_W1TC_REG to disable the output from the GPIO pin.
2. For an open drain output, set the GPIO_PINx_PAD_DRIVER bit in register GPIO_PINx_REG corresponding to GPIO pin X.
3. Configure HP IO MUX register to enable output via HP GPIO matrix. Set IO_MUX_GPIOx_REG corresponding to GPIO pin X as follows:
   - Set the field IO_MUX_GPIOx_MCU_SEL to desired HP IO MUX function corresponding to GPIO pin X. This is Function 1 (GPIO function), numeric value 1, for all pins.
   - Set the IO_MUX_GPIOx_FUN_DRV field to the desired value for output strength (0 ~ 3). The higher the drive strength, the more current can be sourced/sunk from the pin.
   - If using open drain mode, set/clear the IO_MUX_GPIOx_FUN_WPU and IO_MUX_GPIOx_FUN_WPD bits to enable/disable the internal pull-up/pull-down resistors.
4. Enable hysteresis function:
   - GPIO0 ~ GPIO15: set LP_IOMUX_LP_GPIO_HYS[x] (x: 0 ~ 15, corresponding to GPIO0 ~ GPIO15).
   - GPIO16 ~ GPIO47: set HP_SYSTEM_GPIO_O_HYS_LOW[x] (x: 0 ~ 31, corresponding to GPIO16 ~ GPIO47).
   - GPIO48 ~ GPIO54: set HP_SYSTEM_GPIO_O_HYS_HIGH[x] (x: 0 ~ 6, corresponding to GPIO48 ~ GPIO54).

Note:

- The output signal from a single peripheral can be sent to multiple pins simultaneously.
- The output signal can be inverted by setting GPIO_FUNC*x*_OUT_INV_SEL.

9.5.4.2 LP GPIO Matrix

The programming procedure for peripheral input via the LP GPIO matrix resembles that of 9.5.4.1. However, it's worth noting that the control and status registers must use LP GPIO matrix registers. See Section 9.19.4 and Section 9.19.5.

9.6 Direct Input and Output via IO MUX

9.6.1 Overview

Some digital signals (SDMMC, EMAC, etc.) can bypass GPIO matrix for better high-frequency digital performance. In this case, IO MUX is used to connect these pins directly to peripherals. This option is less flexible than routing signals via GPIO matrix, as the IO MUX register for each GPIO pin can only select from a limited number of functions, but high-frequency digital performance can be improved.

ESP32-P4 provides 16 GPIO pins with low power (LP) capabilities. These pins can be controlled by either HP IO MUX or LP IO MUX. If controlled by LP IO MUX, these pins will bypass HP IO MUX and HP GPIO matrix for the use by peripherals in LP system.

When configured as LP GPIOs, the pins can still be controlled by the peripherals in LP system during chip Deep-sleep, and wake up the chip from Deep-sleep.

9.6.2 Functional Description

The pins with LP functions (GPIO0 ~ GPIO15) are controlled by LP_IOMUX_PAD*n*_MUX_SEL (*n*: 0 ~ 15) in register LP_IOMUX_PAD*n*_REG. By default, all bits in these registers are set to 0, routing all input/output signals via HP IO MUX.

If LP_IOMUX_PAD*n*_MUX_SEL is set, then input/output signals are controlled by LP IO MUX. In this mode, LP_IOMUX_PAD*n*_REG is used to control the LP GPIO pins. See 9.15-1 for the LP functions of each LP GPIO pin. Note that LP_IOMUX_PAD*n*_REG applies the LP GPIO pin numbering, not the HP GPIO pin numbering.

9.6.2.1 HP IO MUX

Two fields must be configured in order to bypass HP GPIO matrix for HP peripheral input signals:

1. IO_MUX_GPIO*n*_MCU_SEL for the GPIO pin must be set to the required pin function. For the list of pin functions, please refer to Table 9.14-1.
2. Clear GPIO_SIG*n*_IN_SEL to route the input directly to the peripheral.

To bypass HP GPIO matrix for HP peripheral output signals, IO_MUX_GPIO*n*_MCU_SEL for the GPIO pin must be set to the required pin function.

9.6.2.2 LP IO MUX

Two fields must be configured in order to bypass LP GPIO matrix for LP peripheral input signals:

1. LP_IOMUX_PADn_MUX_SEL for the GPIO pin must be set to the required pin function. For the list of pin functions, please refer to Table 9.15-1.
2. Clear LP_GPIO_GPIO_SIGn_IN_SEL to route the input directly to the peripheral.

To bypass LP GPIO matrix for LP peripheral output signals, LP_IOMUX_PADn_MUX_SEL for the GPIO pin must be set to the required pin function.

> **Note:**
> Not all signals can be directly connected to peripheral via IO MUX. Some input/output signals can only be connected to peripheral via GPIO matrix.

9.7 Analog Functions

9.7.1 Overview

ESP32-P4 provides 34 GPIO pins with analog functions, including 16 LP GPIO pins and 18 HP GPIO pins. See Table 9.16-1.

9.7.2 Analog Functions

When the pin is used for analog purpose, make sure this pin is left floating by configuring registers. By such way, the external analog signal is directly connected to internal analog signal via GPIO pin. The configuration is as follows:

- Clear IO_MUX_GPIOn_FUN_IE, IO_MUX_GPIOn_FUN_WPU, and IO_MUX_GPIOn_FUN_WPD, to set the pin floating.
- Write 1 to the corresponding bit in GPIO_ENABLE_W1TC and GPIO_ENABLE1_W1TC, to clear output enable.

To use these functions listed in Table 9.16-1, please refer to the following related chapters:

- For XTAL_32K related functions, see Chapter 14 *Low-Power Management*.
- For TOUCH related functions, see Chapter 60 *Touch Sensor (TOUCH)*.
- For ADC related functions, see Chapter 62 *ADC Controller (ADC)*.
- For COMP related functions, see Chapter 63 *Analog Voltage Comparator*.

> **Note:**
> GPIO51 ~ GPIO54 each has two different analog functions, which can be used simultaneously.

9.8 Pin Functions in Light-sleep

Pins may provide different functions when ESP32-P4 is in Light-sleep mode. If IO_MUX_GPIO*n*_SLP_SEL in register IO_MUX_GPIO*n*_REG for a GPIO pin is set to 1, a different set of bits will be used to control the pin when the chip is in Light-sleep mode.

Table 9.8-1. Bit Used to Control IO MUX Functions in Light-sleep Mode

IO MUX Function	Normal Execution OR IO_MUX_GPIO*n*_SLP_SEL = 0	Light-sleep Mode AND IO_MUX_GPIO*n*_SLP_SEL = 1
Output Drive Strength	IO_MUX_GPIO*n*_FUN_DRV	IO_MUX_GPIO*n*_MCU_DRV
Pull-up Resistor	IO_MUX_GPIO*n*_FUN_WPU	IO_MUX_GPIO*n*_MCU_WPU
Pull-down Resistor	IO_MUX_GPIO*n*_FUN_WPD	IO_MUX_GPIO*n*_MCU_WPD
Input Enable	IO_MUX_GPIO*n*_FUN_IE	IO_MUX_GPIO*n*_MCU_IE
Output Enable	OE_SEL from GPIO matrix*	IO_MUX_GPIO*n*_MCU_OE

Note:
If IO_MUX_GPIO*n*_SLP_SEL is set to 0, pin functions remain the same in both normal execution and in Light-sleep mode. Please refer to Section 9.5.4.1 for how to enable output in normal execution.

9.9 Pin Hold Feature

Each GPIO pin has an individual hold function controlled by Power Management Unit (PMU) or registers. When the pin is set to hold, the state is latched at that moment and will not change no matter how the internal signals change or how the IO MUX/GPIO configuration is modified. Users can use the hold function for the pins to retain the pin state through a core reset triggered by watchdog time-out or Deep-sleep events.

HP Pins (GPIO16 ~ GPIO54):

The Hold state of each HP pin is controlled by the result of OR operation of the pin's Hold enable signal and the global Hold enable signal.

- Each pin's Hold enable signal is controlled by the following registers:
  - In Deep-sleep mode:
    * LP_SYSTEM_REG_PAD_RTC_HOLD_CTRL0[*n*] (*n* = 16~31), controls the Hold signal of each pin of GPIO16~GPIO31.
    * LP_SYSTEM_REG_PAD_RTC_HOLD_CTRL1[*n*] (*n* = 0~22), controls the Hold signal of each pin of GPIO32~GPIO54.
  - In non-Deep-sleep mode:
    * HP_SYSTEM_GPIO_O_HOLD_LOW[*n*] (*n* = 0~31), controls the Hold signal of each pin of GPIO16~GPIO47.
    * HP_SYSTEM_GPIO_O_HOLD_HIGH[*n*](*n* = 0~6), controls the Hold signal of each pin of GPIO48~GPIO54.

> **Note:**
> In non-Deep-sleep mode, besides the above HP_SYSTEM registers, the LP_SYSTEM registers (LP_SYSTEM_REG_PAD_RTC_HOLD_CTRL0/1) can also control the Hold function for corresponding pins, with the same effect as the HP_SYSTEM registers.

- PMU_TIE_HIGH_HP_PAD_HOLD_ALL, controls the global Hold signal of all HP pins.

To use this feature, follow the steps below:

- To maintain the pin's input/output status in Deep-sleep, set LP_SYSTEM_REG_PAD_RTC_HOLD_CTRL0[*n*] (*n* = 16~31) or LP_SYSTEM_REG_PAD_RTC_HOLD_CTRL1[*n*] (*n* = 0~22) before powering down. To disable the hold function after waking up, clear both registers.
- To maintain the pin's input/output status in non-Deep-sleep mode, set HP_SYSTEM_GPIO_O_HOLD_LOW[*n*] (*n* = 0~31) or HP_SYSTEM_GPIO_O_HOLD_HIGH[*n*] (*n* = 0~6) before powering down. To disable the hold function after waking up, clear both registers. You can also use the LP_SYSTEM registers (LP_SYSTEM_REG_PAD_RTC_HOLD_CTRL0/1) as described above.
- Alternatively, set PMU_TIE_HIGH_HP_PAD_HOLD_ALL to maintain the input/output status of all HP pins and set PMU_TIE_LOW_HP_PAD_HOLD_ALL to disable the hold function for all HP pins.
- Alternatively, configure the PMU task to hold the HP pins before Deep-sleep. In Deep-sleep, the HP pins would be hold automatically by the PMU.

LP Pins (GPIO0 ~ GPIO15):

The Hold state of each LP pin is controlled by the result of OR operation of the pin's Hold enable signal and the global Hold enable signal.

- Each pin's Hold enable signal is controlled by the following registers:
  - In Deep-sleep mode, LP_SYSTEM_REG_PAD_RTC_HOLD_CTRL0[*n*] (*n* = 0~15) controls the Hold signal of each pin of GPIO0~GPIO15.
  - In non-Deep-sleep mode, LP_IOMUX_LP_PAD_HOLD_REG[*n*] (*n* = 0~15) controls the Hold signal of each pin of GPIO0~GPIO15.

  > **Note:**
  > In non-Deep-sleep mode, besides the above LP_IOMUX register, the LP_SYSTEM register (LP_SYSTEM_REG_PAD_RTC_HOLD_CTRL0) can also control the Hold function for corresponding pins, with the same effect as the LP_IOMUX register.

- PMU_TIE_HIGH_LP_PAD_HOLD_ALL controls the global Hold signal of all LP pins.

To use this feature, follow the steps below:

- To maintain the pin's input/output status in Deep-sleep, set LP_SYSTEM_REG_PAD_RTC_HOLD_CTRL0[*n*] (*n* = 0~15) to hold the value of GPIO*n*, or clear LP_SYSTEM_REG_PAD_RTC_HOLD_CTRL0[*n*] (*n* = 0~15) to disable the hold function of GPIO*n*.
- To maintain the pin's input/output status in non-Deep-sleep mode, set LP_IOMUX_LP_PAD_HOLD_REG[*n*] (*n* = 0~15) to hold the value of GPIO*n*, or clear

LP_IOMUX_LP_PAD_HOLD_REG[n] (n = 0∼15) to disable the hold function of GPIOn. You can also use the LP_SYSTEM register (LP_SYSTEM_REG_PAD_RTC_HOLD_CTRL0) as described above.

- Alternatively, set PMU_TIE_HIGH_LP_PAD_HOLD_ALL to hold the values of all LP pins, and set PMU_TIE_LOW_LP_PAD_HOLD_ALL to disable the hold function of all LP pins.
- Alternatively, configure the PMU task to hold LP pins before Deep-sleep. In Deep-sleep, LP pins will be hold automatically by the PMU.

9.10 Hysteresis Characteristics of GPIO Pins

Each GPIO pin has hysteresis functionality. When hysteresis is not enabled, as shown in Figure 9.10-1, the level flip of the signal (C) input to the chip from the PAD has only one threshold (Vt, about 1.7 V). When the voltage on the PAD is higher than Vt, the level on the C is high. Otherwise, it is low. However, noise on the PAD may affect the signal on C.

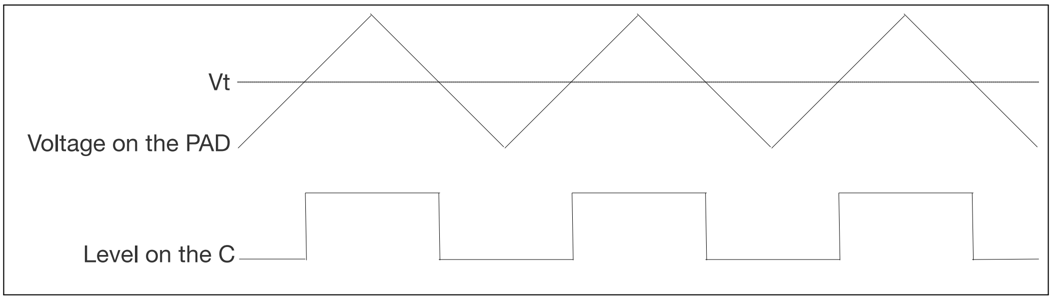

Figure 9.10-1. Example of Level Flip on the Chip Pad — Hysteresis Function Not Enabled

When hysteresis is enabled, as shown in Figure 9.10-2, the level flip of C has two thresholds, high-level threshold (Vth, about 1.7 V) and low-level threshold (Vtl, about 1.4 V). When the voltage of pad goes from low to high, if the voltage is higher than Vth, the level of C is high. When the voltage of PAD goes from high to low, if the voltage is lower than Vtl, the level of C is low. When the voltage of PAD is between Vth and Vtl, the level of C does not change. The hysteresis function plays an anti-interference role by mitigating the impact of noise, consequently reducing the level flip time of C.

To enable the hysteresis function, follow the steps below:

- Set LP_IOMUX_LP_GPIO_HYS[n] (n ranges from 0 ~ 15, corresponding to GPIO0 ~ GPIO15), HP_SYS_GPIO0_HYS_LOW[n] (n ranges from 0 ~ 31, corresponding to GPIO16 ~ GPIO47) or HP_SYS_GPIO0_HYS_HIGH[n] (n ranges from 0 ~ 6, corresponding to GPIO48 ~ GPIO54) to 1 to enable the hysteresis function for GPIOn.
- Set LP_IOMUX_LP_GPIO_HYS[n] (n ranges from 0 ~ 15, corresponding to GPIO0 ~ GPIO15), HP_SYS_GPIO0_HYS_LOW[n] (n ranges from 0 ~ 31, corresponding to GPIO16 ~ GPIO47) or HP_SYS_GPIO0_HYS_HIGH[n] (n ranges from 0 ~ 6, corresponding to GPIO48 ~ GPIO54) to 0 to disable the hysteresis function for GPIOn.

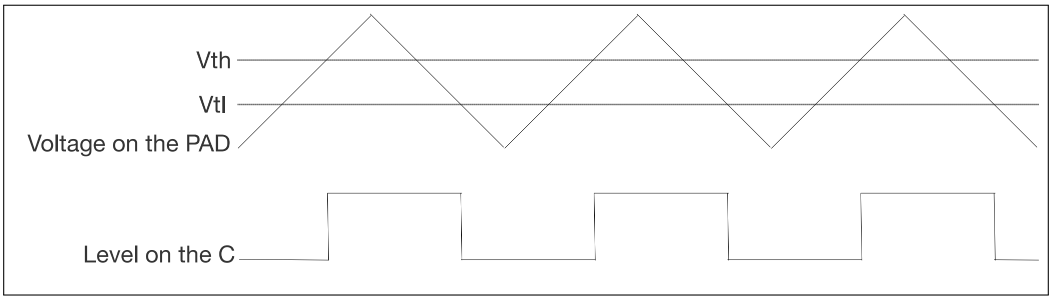

Figure 9.10-2. Example of Level Flip on the Chip Pad — Hysteresis Function Enabled

9.11 Power Supplies and Management of GPIO Pins

9.11.1 Power Supplies of GPIO Pins

For more information on the power supply for GPIO pins, please refer to Pin Definition in *ESP32-P4 Datasheet*. All the pins can be used to wake up the chip from Light-sleep, but only the pins (GPIO0 ~ GPIO15) in VDD_LP domain can be used to wake up the chip from Deep-sleep.

9.11.2 Power Supply Management

Each ESP32-P4 pin is connected to one of the following power domains.

- VDD_LP: the input power supply for LP GPIO pins
- VDD_IO_0, VDD_FLASHIO, VDD_IO_4 ~ VDD_IO_6: the input power supply for HP GPIO pins

9.12 HP Peripheral Signal List

Table 9.12-1 shows the peripheral input/output signals via HP GPIO matrix.

Please pay attention to the configuration of the bit GPIO_FUNC*n*_OE_SEL:

- GPIO_FUNC*n*_OE_SEL = 1: the output enable is controlled by the corresponding bit *n* of GPIO_ENABLE/ENABLE1_REG:
  - GPIO_ENABLE/ENABLE1_REG = 0: output is disabled.
  - GPIO_ENABLE/ENABLE1_REG = 1: output is enabled.
- GPIO_FUNC*n*_OE_SEL = 0: use the output enable signal from peripheral, for example spi2_dqs_pad_oe in the column "Output enable signal when GPIO_FUNC*n*_OE_SEL = 0" of Table 9.12-1. Note that the signals such as spi2_dqs_pad_oe can be 1 (1'd1) or 0 (1'd0), depending on the configuration of corresponding peripherals. If it's 1'd1 in column "Output enable signal when GPIO_FUNC*n*_OE_SEL = 0", it indicates that once GPIO_FUNC*n*_OE_SEL is cleared, the output signal is always enabled by default.

Note:
Signals are numbered consecutively, but not all signals are valid.

- Only the signals with a name assigned in the column "Input signal" in Table 9.12-1 are valid input signals.
- Only the signals with a name assigned in the column "Output signal" in Table 9.12-1 are valid output signals.

Table 9.12-1. Peripheral Signals via HP GPIO Matrix

Signal No.	Input Signal	Default Value	Direct Input via HP IO MUX	Output Signal	Output Enable Signal when GPIO_FUNCn_OE_SEL = 0	Direct Output via HP IO MUX
0	-	-	-	sd_card_cclk_2_pad_out	1'd1	no
1	sd_card_ccmd_2_pad_in	1	no	sd_card_ccmd_2_pad_out	sd_card_ccmd_2_pad_oe	no
2	sd_card_cdata0_2_pad_in	1	no	sd_card_cdata0_2_pad_out	sd_card_cdata0_2_pad_oe	no
3	sd_card_cdata1_2_pad_in	1	no	sd_card_cdata1_2_pad_out	sd_card_cdata1_2_pad_oe	no
4	sd_card_cdata2_2_pad_in	1	no	sd_card_cdata2_2_pad_out	sd_card_cdata2_2_pad_oe	no
5	sd_card_cdata3_2_pad_in	1	no	sd_card_cdata3_2_pad_out	sd_card_cdata3_2_pad_oe	no
6	sd_card_cdata4_2_pad_in	1	no	sd_card_cdata4_2_pad_out	sd_card_cdata4_2_pad_oe	no
7	sd_card_cdata5_2_pad_in	1	no	sd_card_cdata5_2_pad_out	sd_card_cdata5_2_pad_oe	no
8	sd_card_cdata6_2_pad_in	1	no	sd_card_cdata6_2_pad_out	sd_card_cdata6_2_pad_oe	no
9	sd_card_cdata7_2_pad_in	1	no	sd_card_cdata7_2_pad_out	sd_card_cdata7_2_pad_oe	no
10	uart0_rxd_pad_in	0	yes	uart0_txd_pad_out	1'd1	yes
11	uart0_cts_pad_in	0	yes	uart0_rts_pad_out	1'd1	yes
12	uart0_dsr_pad_in	0	no	uart0_dtr_pad_out	1'd1	no
13	uart1_rxd_pad_in	0	yes	uart1_txd_pad_out	1'd1	yes
14	uart1_cts_pad_in	0	yes	uart1_rts_pad_out	1'd1	yes
15	uart1_dsr_pad_in	0	no	uart1_dtr_pad_out	1'd1	no
16	uart2_rxd_pad_in	0	no	uart2_txd_pad_out	1'd1	no
17	uart2_cts_pad_in	0	no	uart2_rts_pad_out	1'd1	no
18	uart2_dsr_pad_in	0	no	uart2_dtr_pad_out	1'd1	no
19	uart3_rxd_pad_in	0	no	uart3_txd_pad_out	1'd1	no
20	uart3_cts_pad_in	0	no	uart3_rts_pad_out	1'd1	no
21	uart3_dsr_pad_in	0	no	uart3_dtr_pad_out	1'd1	no
22	uart4_rxd_pad_in	0	no	uart4_txd_pad_out	1'd1	no
23	uart4_cts_pad_in	0	no	uart4_rts_pad_out	1'd1	no
24	uart4_dsr_pad_in	0	no	uart4_dtr_pad_out	1'd1	no
25	i2s0_o_bck_pad_in	0	no	i2s0_o_bck_pad_out	1'd1	no
26	i2s0_mclk_pad_in	0	no	i2s0_mclk_pad_out	1'd1	no

Signal No.	Input Signal	Default Value	Direct Input via HP IO MUX	Output Signal	Output Enable Signal when GPIO_FUNCn_OE_SEL = 0	Direct Output via HP IO MUX
27	i2s0_o_ws_pad_in	0	no	i2s0_o_ws_pad_out	1'd1	no
28	i2s0_i_sd_pad_in	0	no	i2s0_o_sd_pad_out	1'd1	no
29	i2s0_i_bck_pad_in	0	no	i2s0_i_bck_pad_out	1'd1	no
30	i2s0_i_ws_pad_in	0	no	i2s0_i_ws_pad_out	1'd1	no
31	i2s1_o_bck_pad_in	0	no	i2s1_o_bck_pad_out	1'd1	no
32	i2s1_mclk_pad_in	0	no	i2s1_mclk_pad_out	1'd1	no
33	i2s1_o_ws_pad_in	0	no	i2s1_o_ws_pad_out	1'd1	no
34	i2s1_i_sd_pad_in	0	no	i2s1_o_sd_pad_out	1'd1	no
35	i2s1_i_bck_pad_in	0	no	i2s1_i_bck_pad_out	1'd1	no
36	i2s1_i_ws_pad_in	0	no	i2s1_i_ws_pad_out	1'd1	no
37	i2s2_o_bck_pad_in	0	no	i2s2_o_bck_pad_out	1'd1	no
38	i2s2_mclk_pad_in	0	no	i2s2_mclk_pad_out	1'd1	no
39	i2s2_o_ws_pad_in	0	no	i2s2_o_ws_pad_out	1'd1	no
40	i2s2_i_sd_pad_in	0	no	i2s2_o_sd_pad_out	1'd1	no
41	i2s2_i_bck_pad_in	0	no	i2s2_i_bck_pad_out	1'd1	no
42	i2s2_i_ws_pad_in	0	no	i2s2_i_ws_pad_out	1'd1	no
43	i2s0_i_sd1_pad_in	0	no	i2s0_o_sd1_pad_out	1'd1	no
44	i2s0_i_sd2_pad_in	0	no	spi2_dqs_pad_out	spi2_dqs_pad_oe	yes
45	i2s0_i_sd3_pad_in	0	no	spi3_cs2_pad_out	spi3_cs2_pad_oe	no
46	-	-	-	spi3_cs1_pad_out	spi3_cs1_pad_oe	no
47	spi3_ck_pad_in	0	no	spi3_ck_pad_out	spi3_ck_pad_oe	no
48	spi3_q_pad_in	0	no	spi3_qo_pad_out	spi3_q_pad_oe	no
49	spi3_d_pad_in	0	no	spi3_d_pad_out	spi3_d_pad_oe	no
50	spi3_hold_pad_in	0	no	spi3_hold_pad_out	spi3_hold_pad_oe	no
51	spi3_wp_pad_in	0	no	spi3_wp_pad_out	spi3_wp_pad_oe	no
52	spi3_cs_pad_in	0	no	spi3_cs_pad_out	spi3_cs_pad_oe	no
53	spi2_ck_pad_in	0	yes	spi2_ck_pad_out	spi2_ck_pad_oe	yes
54	spi2_q_pad_in	0	yes	spi2_q_pad_out	spi2_q_pad_oe	yes

Signal No.	Input Signal	Default Value	Direct Input via HP IO MUX	Output Signal	Output Enable Signal when GPIO_FUNCn_OE_SEL = 0	Direct Output via HP IO MUX
55	spi2_d_pad_in	0	yes	spi2_d_pad_out	spi2_d_pad_oe	yes
56	spi2_hold_pad_in	0	yes	spi2_hold_pad_out	spi2_hold_pad_oe	yes
57	spi2_wp_pad_in	0	yes	spi2_wp_pad_out	spi2_wp_pad_oe	yes
58	spi2_io4_pad_in	0	yes	spi2_io4_pad_out	spi2_io4_pad_oe	yes
59	spi2_io5_pad_in	0	yes	spi2_io5_pad_out	spi2_io5_pad_oe	yes
60	spi2_io6_pad_in	0	yes	spi2_io6_pad_out	spi2_io6_pad_oe	yes
61	spi2_io7_pad_in	0	yes	spi2_io7_pad_out	spi2_io7_pad_oe	yes
62	spi2_cs_pad_in	0	yes	spi2_cs_pad_out	spi2_cs_pad_oe	yes
63	pcnt_rst_pad_in0	0	no	spi2_cs1_pad_out	spi2_cs1_pad_oe	no
64	pcnt_rst_pad_in1	0	no	spi2_cs2_pad_out	spi2_cs2_pad_oe	no
65	pcnt_rst_pad_in2	0	no	spi2_cs3_pad_out	spi2_cs3_pad_oe	no
66	pcnt_rst_pad_in3	0	no	spi2_cs4_pad_out	spi2_cs4_pad_oe	no
67	-	-	-	spi2_cs5_pad_out	spi2_cs5_pad_oe	no
68	i2c0_scl_pad_in	1	no	i2c0_scl_pad_out	i2c0_scl_oe_pad_out	no
69	i2c0_sda_pad_in	1	no	i2c0_sda_pad_out	i2c0_sda_oe_pad_out	no
70	i2c1_scl_pad_in	1	no	i2c1_scl_pad_out	i2c1_scl_oe_pad_out	no
71	i2c1_sda_pad_in	1	no	i2c1_sda_pad_out	i2c1_sda_oe_pad_out	no
72	-	-	-	gpio_sd0_out	1'd1	no
73	-	-	-	gpio_sd1_out	1'd1	no
74	uart0_slp_clk_pad_in	0	no	gpio_sd2_out	1'd1	no
75	uart1_slp_clk_pad_in	0	no	gpio_sd3_out	1'd1	no
76	uart2_slp_clk_pad_in	0	no	gpio_sd4_out	1'd1	no
77	uart3_slp_clk_pad_in	0	no	gpio_sd5_out	1'd1	no
78	uart4_slp_clk_pad_in	0	no	gpio_sd6_out	1'd1	no
79	-	-	-	gpio_sd7_out	1'd1	no
80	twai0_rx_pad_in	1	no	twai0_tx_pad_out	1'd1	no
81	-	-	-	twai0_bus_off_on_pad_out	1'd1	no
82	-	-	-	twai0_clkout_pad_out	1'd1	no

Signal No.	Input Signal	Default Value	Direct Input via HP IO MUX	Output Signal	Output Enable Signal when GPIO_FUNCn_OE_SEL = 0	Direct Output via HP IO MUX
83	twai1_rx_pad_in	1	no	twai1_tx_pad_out	1'd1	no
84	-	-	-	twai1_bus_off_on_pad_out	1'd1	no
85	-	-	-	twai1_clkout_pad_out	1'd1	no
86	twai2_rx_pad_in	1	no	twai2_tx_pad_out	1'd1	no
87	-	-	-	twai2_bus_off_on_pad_out	1'd1	no
88	-	-	-	twai2_clkout_pad_out	1'd1	no
89	pwm0_sync0_pad_in	0	no	pwm0_ch0_a_pad_out	1'd1	no
90	pwm0_sync1_pad_in	0	no	pwm0_ch0_b_pad_out	1'd1	no
91	pwm0_sync2_pad_in	0	no	pwm0_ch1_a_pad_out	1'd1	no
92	pwm0_f0_pad_in	0	no	pwm0_ch1_b_pad_out	1'd1	no
93	pwm0_f1_pad_in	0	no	pwm0_ch2_a_pad_out	1'd1	no
94	pwm0_f2_pad_in	0	no	pwm0_ch2_b_pad_out	1'd1	no
95	pwm0_cap0_pad_in	0	no	pwm1_ch0_a_pad_out	1'd1	no
96	pwm0_cap1_pad_in	0	no	pwm1_ch0_b_pad_out	1'd1	no
97	pwm0_cap2_pad_in	0	no	pwm1_ch1_a_pad_out	1'd1	no
98	pwm1_sync0_pad_in	0	no	pwm1_ch1_b_pad_out	1'd1	no
99	pwm1_sync1_pad_in	0	no	pwm1_ch2_a_pad_out	1'd1	no
100	pwm1_sync2_pad_in	0	no	pwm1_ch2_b_pad_out	1'd1	no
101	pwm1_f0_pad_in	0	no	-	-	-
102	pwm1_f1_pad_in	0	no	-	-	-
103	pwm1_f2_pad_in	0	no	-	-	-
104	pwm1_cap0_pad_in	0	no	-	-	-
105	pwm1_cap1_pad_in	0	no	twai0_standby_pad_out	1'd1	no
106	pwm1_cap2_pad_in	0	no	twai1_standby_pad_out	1'd1	no
107	gmii_mdi_pad_in	0	no	twai2_standby_pad_out	1'd1	no
108	gmac_phy_col_pad_in	0	no	gmii_mdc_pad_out	1'd1	no
109	gmac_phy_crs_pad_in	0	no	gmii_mdo_pad_out	gmii_mdo_oe_pad_out	no
110	usb_otg11_iddig_pad_in	0	no	usb_srp_dischrgvbus_pad_out	1'd1	no

Signal No.	Input Signal	Default Value	Direct Input via HP IO MUX	Output Signal	Output Enable Signal when GPIO_FUNCn_OE_SEL = 0	Direct Output via HP IO MUX
111	usb_otg11_avalid_pad_in	0	no	usb_otg11_idpullup_pad_out	1'd1	no
112	usb_srp_bvalid_pad_in	0	no	usb_otg11_dppulldown_pad_out	1'd1	no
113	usb_otg11_vbusvalid_pad_in	0	no	usb_otg11_dmpulldown_pad_out	1'd1	no
114	usb_srp_sessend_pad_in	0	no	usb_otg11_drvvbus_pad_out	1'd1	no
115	-	-	-	usb_srp_chrgvbus_pad_out	1'd1	no
116	-	-	-	-	-	-
117	ulpi_clk_pad_in	0	no	-	-	-
118	usb_hsphy_refclk_in	0	no	-	-	-
119	-	-	-	-	-	-
120	-	-	-	-	-	-
121	-	-	-	-	-	-
122	-	-	-	-	-	-
123	-	-	-	-	-	-
124	-	-	-	-	-	-
125	-	-	-	-	-	-
126	sd_card_detect_n_1_pad_in	0	no	ledc_ls_sig_out_pad_out0	1'd1	no
127	sd_card_detect_n_2_pad_in	0	no	ledc_ls_sig_out_pad_out1	1'd1	no
128	sd_card_int_n_1_pad_in	1	no	ledc_ls_sig_out_pad_out2	1'd1	no
129	sd_card_int_n_2_pad_in	1	no	ledc_ls_sig_out_pad_out3	1'd1	no
130	sd_card_write_prt_1_pad_in	0	no	ledc_ls_sig_out_pad_out4	1'd1	no
131	sd_card_write_prt_2_pad_in	0	no	ledc_ls_sig_out_pad_out5	1'd1	no
132	sd_data_strobe_1_pad_in	0	no	ledc_ls_sig_out_pad_out6	1'd1	no
133	sd_data_strobe_2_pad_in	0	no	ledc_ls_sig_out_pad_out7	1'd1	no
134	i3c_mst_scl_pad_in	1	no	i3c_mst_scl_pad_out	i3c_mst_scl_oe_pad_out	no
135	i3c_mst_sda_pad_in	1	no	i3c_mst_sda_pad_out	i3c_mst_sda_oe_pad_out	no
136	i3c_slv_scl_pad_in	1	no	i3c_slv_scl_pad_out	i3c_slv_scl_oe_pad_out	no
137	i3c_slv_sda_pad_in	1	no	i3c_slv_sda_pad_out	i3c_slv_sda_oe_pad_out	no

GoBack

Signal No.	Input Signal	Default Value	Direct Input via HP IO MUX	Output Signal	Output Enable Signal when GPIO_FUNCn_OE_SEL = 0	Direct Output via HP IO MUX
138	-	-	-	i3c_mst_scl_pullup_en_pad_out	1'd1	no
139	-	-	-	i3c_mst_sda_pullup_en_pad_out	1'd1	no
140	usb_jtag_tdo_bridge_pad_in	0	no	usb_jtag_tdi_bridge_pad_out	1'd1	no
141	pcnt_sig_ch0_pad_in0	0	no	usb_jtag_tms_bridge_pad_out	1'd1	no
142	pcnt_sig_ch0_pad_in1	0	no	usb_jtag_tck_bridge_pad_out	1'd1	no
143	pcnt_sig_ch0_pad_in2	0	no	usb_jtag_trst_bridge_pad_out	1'd1	no
144	pcnt_sig_ch0_pad_in3	0	no	lcd_cs_pad_out	1'd1	no
145	pcnt_sig_ch1_pad_in0	0	no	lcd_dc_pad_out	1'd1	no
146	pcnt_sig_ch1_pad_in1	0	no	sd_rst_n_1_pad_out	1'd1	no
147	pcnt_sig_ch1_pad_in2	0	no	sd_rst_n_2_pad_out	1'd1	no
148	pcnt_sig_ch1_pad_in3	0	no	sd_ccmd_od_pullup_en_n_pad_out	1'd1	no
149	pcnt_ctrl_ch0_pad_in0	0	no	lcd_pclk_pad_out	1'd1	no
150	pcnt_ctrl_ch0_pad_in1	0	no	cam_clk_pad_out	1'd1	no
151	pcnt_ctrl_ch0_pad_in2	0	no	lcd_h_enable_pad_out	1'd1	no
152	pcnt_ctrl_ch0_pad_in3	0	no	lcd_h_sync_pad_out	1'd1	no
153	pcnt_ctrl_ch1_pad_in0	0	no	lcd_v_sync_pad_out	1'd1	no
154	pcnt_ctrl_ch1_pad_in1	0	no	lcd_data_out_pad_out0	1'd1	no
155	pcnt_ctrl_ch1_pad_in2	0	no	lcd_data_out_pad_out1	1'd1	no
156	pcnt_ctrl_ch1_pad_in3	0	no	lcd_data_out_pad_out2	1'd1	no
157	-	-	-	lcd_data_out_pad_out3	1'd1	no
158	cam_pclk_pad_in	0	no	lcd_data_out_pad_out4	1'd1	no
159	cam_h_enable_pad_in	0	no	lcd_data_out_pad_out5	1'd1	no
160	cam_h_sync_pad_in	0	no	lcd_data_out_pad_out6	1'd1	no
161	cam_v_sync_pad_in	0	no	lcd_data_out_pad_out7	1'd1	no
162	cam_data_in_pad_in0	0	no	lcd_data_out_pad_out8	1'd1	no
163	cam_data_in_pad_in1	0	no	lcd_data_out_pad_out9	1'd1	no

Signal No.	Input Signal	Default Value	Direct Input via HP IO MUX	Output Signal	Output Enable Signal when GPIO_FUNCn_OE_SEL = 0	Direct Output via HP IO MUX
164	cam_data_in_pad_in2	0	no	lcd_data_out_pad_out10	1'd1	no
165	cam_data_in_pad_in3	0	no	lcd_data_out_pad_out11	1'd1	no
166	cam_data_in_pad_in4	0	no	lcd_data_out_pad_out12	1'd1	no
167	cam_data_in_pad_in5	0	no	lcd_data_out_pad_out13	1'd1	no
168	cam_data_in_pad_in6	0	no	lcd_data_out_pad_out14	1'd1	no
169	cam_data_in_pad_in7	0	no	lcd_data_out_pad_out15	1'd1	no
170	cam_data_in_pad_in8	0	no	lcd_data_out_pad_out16	1'd1	no
171	cam_data_in_pad_in9	0	no	lcd_data_out_pad_out17	1'd1	no
172	cam_data_in_pad_in10	0	no	lcd_data_out_pad_out18	1'd1	no
173	cam_data_in_pad_in11	0	no	lcd_data_out_pad_out19	1'd1	no
174	cam_data_in_pad_in12	0	no	lcd_data_out_pad_out20	1'd1	no
175	cam_data_in_pad_in13	0	no	lcd_data_out_pad_out21	1'd1	no
176	cam_data_in_pad_in14	0	no	lcd_data_out_pad_out22	1'd1	no
177	cam_data_in_pad_in15	0	no	lcd_data_out_pad_out23	1'd1	no
178	gmac_phy_rxdv_pad_in	0	yes	gmac_phy_txen_pad_out	1'd1	yes
179	gmac_phy_rxd0_pad_in	0	yes	gmac_phy_txd0_pad_out	1'd1	yes
180	gmac_phy_rxd1_pad_in	0	yes	gmac_phy_txd1_pad_out	1'd1	yes
181	gmac_phy_rxd2_pad_in	0	no	gmac_phy_txd2_pad_out	1'd1	no
182	gmac_phy_rxd3_pad_in	0	no	gmac_phy_txd3_pad_out	1'd1	no
183	gmac_phy_rxer_pad_in	0	yes	gmac_phy_txer_pad_out	1'd1	yes
184	gmac_rx_clk_pad_in	0	no	-	-	-
185	gmac_tx_clk_pad_in	0	no	-	-	-
186	parlio_rx_clk_pad_in	0	no	parlio_rx_clk_pad_out	1'd1	no
187	parlio_tx_clk_pad_in	0	no	parlio_tx_clk_pad_out	1'd1	no
188	parlio_rx_data0_pad_in	0	no	parlio_tx_data0_pad_out	1'd1	no
189	parlio_rx_data1_pad_in	0	no	parlio_tx_data1_pad_out	1'd1	no
190	parlio_rx_data2_pad_in	0	no	parlio_tx_data2_pad_out	1'd1	no
191	parlio_rx_data3_pad_in	0	no	parlio_tx_data3_pad_out	1'd1	no

Signal No.	Input Signal	Default Value	Direct Input via HP IO MUX	Output Signal	Output Enable Signal when GPIO_FUNCn_OE_SEL = 0	Direct Output via HP IO MUX
192	parlio_rx_data4_pad_in	0	no	parlio_tx_data4_pad_out	1'd1	no
193	parlio_rx_data5_pad_in	0	no	parlio_tx_data5_pad_out	1'd1	no
194	parlio_rx_data6_pad_in	0	no	parlio_tx_data6_pad_out	1'd1	no
195	parlio_rx_data7_pad_in	0	no	parlio_tx_data7_pad_out	1'd1	no
196	parlio_rx_data8_pad_in	0	no	parlio_tx_data8_pad_out	1'd1	no
197	parlio_rx_data9_pad_in	0	no	parlio_tx_data9_pad_out	1'd1	no
198	parlio_rx_data10_pad_in	0	no	parlio_tx_data10_pad_out	1'd1	no
199	parlio_rx_data11_pad_in	0	no	parlio_tx_data11_pad_out	1'd1	no
200	parlio_rx_data12_pad_in	0	no	parlio_tx_data12_pad_out	1'd1	no
201	parlio_rx_data13_pad_in	0	no	parlio_tx_data13_pad_out	1'd1	no
202	parlio_rx_data14_pad_in	0	no	parlio_tx_data14_pad_out	1'd1	no
203	parlio_rx_data15_pad_in	0	no	parlio_tx_data15_pad_out	1'd1	no
204	-	-	-	-	-	-
205	-	-	-	-	-	-
206	-	-	-	-	-	-
207	-	-	-	-	-	-
208	-	-	-	-	-	-
209	-	-	-	-	-	-
210	-	-	-	-	-	-
211	-	-	-	-	-	-
212	-	-	-	constant0_pad_out	1'd1	no
213	-	-	-	constant1_pad_out	1'd1	no
214	core_gpio_in_pad_in0	0	no	core_gpio_out_pad_out0	core_gpio_oe_pad_out0	no
215	core_gpio_in_pad_in1	0	no	core_gpio_out_pad_out1	core_gpio_oe_pad_out1	no
216	core_gpio_in_pad_in2	0	no	core_gpio_out_pad_out2	core_gpio_oe_pad_out2	no
217	core_gpio_in_pad_in3	0	no	core_gpio_out_pad_out3	core_gpio_oe_pad_out3	no
218	core_gpio_in_pad_in4	0	no	core_gpio_out_pad_out4	core_gpio_oe_pad_out4	no
219	core_gpio_in_pad_in5	0	no	core_gpio_out_pad_out5	core_gpio_oe_pad_out5	no

Signal No.	Input Signal	Default Value	Direct Input via HP IO MUX	Output Signal	Output Enable Signal when GPIO_FUNCn_OE_SEL = 0	Direct Output via HP IO MUX
220	core_gpio_in_pad_in6	0	no	core_gpio_out_pad_out6	core_gpio_oe_pad_out6	no
221	core_gpio_in_pad_in7	0	no	core_gpio_out_pad_out7	core_gpio_oe_pad_out7	no
222	core_gpio_in_pad_in8	0	no	core_gpio_out_pad_out8	core_gpio_oe_pad_out8	no
223	core_gpio_in_pad_in9	0	no	core_gpio_out_pad_out9	core_gpio_oe_pad_out9	no
224	core_gpio_in_pad_in10	0	no	core_gpio_out_pad_out10	core_gpio_oe_pad_out10	no
225	core_gpio_in_pad_in11	0	no	core_gpio_out_pad_out11	core_gpio_oe_pad_out11	no
226	core_gpio_in_pad_in12	0	no	core_gpio_out_pad_out12	core_gpio_oe_pad_out12	no
227	core_gpio_in_pad_in13	0	no	core_gpio_out_pad_out13	core_gpio_oe_pad_out13	no
228	core_gpio_in_pad_in14	0	no	core_gpio_out_pad_out14	core_gpio_oe_pad_out14	no
229	core_gpio_in_pad_in15	0	no	core_gpio_out_pad_out15	core_gpio_oe_pad_out15	no
230	core_gpio_in_pad_in16	0	no	core_gpio_out_pad_out16	core_gpio_oe_pad_out16	no
231	core_gpio_in_pad_in17	0	no	core_gpio_out_pad_out17	core_gpio_oe_pad_out17	no
232	core_gpio_in_pad_in18	0	no	core_gpio_out_pad_out18	core_gpio_oe_pad_out18	no
233	core_gpio_in_pad_in19	0	no	core_gpio_out_pad_out19	core_gpio_oe_pad_out19	no
234	core_gpio_in_pad_in20	0	no	core_gpio_out_pad_out20	core_gpio_oe_pad_out20	no
235	core_gpio_in_pad_in21	0	no	core_gpio_out_pad_out21	core_gpio_oe_pad_out21	no
236	core_gpio_in_pad_in22	0	no	core_gpio_out_pad_out22	core_gpio_oe_pad_out22	no
237	core_gpio_in_pad_in23	0	no	core_gpio_out_pad_out23	core_gpio_oe_pad_out23	no
238	core_gpio_in_pad_in24	0	no	core_gpio_out_pad_out24	core_gpio_oe_pad_out24	no
239	core_gpio_in_pad_in25	0	no	core_gpio_out_pad_out25	core_gpio_oe_pad_out25	no
240	core_gpio_in_pad_in26	0	no	core_gpio_out_pad_out26	core_gpio_oe_pad_out26	no
241	core_gpio_in_pad_in27	0	no	core_gpio_out_pad_out27	core_gpio_oe_pad_out27	no
242	core_gpio_in_pad_in28	0	no	parlio_tx_cs_pad_out	1'd1	no
243	core_gpio_in_pad_in29	0	no	emac_ptp_pps_pad_out	1'd1	no
244	core_gpio_in_pad_in30	0	no	ana_comp0_out	1'd1	no
245	core_gpio_in_pad_in31	0	no	ana_comp1_out	1'd1	no
246	rmt_sig_pad_in0	0	no	rmt_sig_pad_out0	1'd1	no
247	rmt_sig_pad_in1	0	no	rmt_sig_pad_out1	1'd1	no

Signal No.	Input Signal	Default Value	Direct Input via HP IO MUX	Output Signal	Output Enable Signal when GPIO_FUNCn_OE_SEL = 0	Direct Output via HP IO MUX
248	rmt_sig_pad_in2	0	no	rmt_sig_pad_out2	1'd1	no
249	rmt_sig_pad_in3	0	no	rmt_sig_pad_out3	1'd1	no
250	sig_in_func250	0	no	sig_in_func250	1'd1	no
251	sig_in_func251	0	no	sig_in_func251	1'd1	no
252	sig_in_func252	0	no	sig_in_func252	1'd1	no
253	sig_in_func253	0	no	sig_in_func253	1'd1	no
254	sig_in_func254	0	no	sig_in_func254	1'd1	no
255	sig_in_func255	0	no	sig_in_func255	1'd1	no

9.13 LP Peripheral Signal List

Table 9.13-1 shows the peripheral input/output signals via LP GPIO matrix.

Please pay attention to the configuration of the bit LP_GPIO_FUNC*n*_OE_SEL:

- LP_GPIO_FUNC*n*_OE_SEL = 1: the output enable is controlled by the corresponding bit*n* of LP_GPIO_ENABLE_REG:
  - LP_GPIO_ENABLE_REG = 0: output is disabled.
  - LP_GPIO_ENABLE_REG = 1: output is enabled.
- LP_GPIO_FUNC*n*_OE_SEL = 0: use the output enable signal from peripheral, for example lp_spi_clk_pad_oe in the column "Output enable signal when LP_GPIO_FUNC*n*_OE_SEL = 0" of Table 9.13-1. Note that the signals such as lp_spi_clk_pad_oe can be 1 (1'd1) or 0 (1'd0), depending on the configuration of corresponding peripherals. If it's 1'd1 in column "Output enable signal when LP_GPIO_FUNC*n*_OE_SEL = 0", it indicates that once LP_GPIO_FUNC*n*_OE_SEL is cleared, the output signal is always enabled by default.

Note:

Signals are numbered consecutively, but not all signals are valid.

- Only the signals with a name assigned in the column "Input signal" in Table 9.13-1 are valid input signals.
- Only the signals with a name assigned in the column "Output signal" in Table 9.13-1 are valid output signals.

Table 9.13-1. Peripheral Signals via LP GPIO Matrix

Signal No.	Input Signal	Default value	Direct Input via LP IO MUX	Output Signal	Output enable signal when LP_GPIO_FUNCn_OE_SEL = 0	Direct Output via LP IO MUX
0	lp_i2c_scl_pad_in	1	no	lp_i2c_scl_pad_out	lp_i2c_scl_pad_oe	no
1	lp_i2c_sda_pad_in	1	no	lp_i2c_sda_pad_out	lp_i2c_sda_pad_oe	no
2	lp_uart_rxd_pad_in	0	yes	lp_uart_txd_pad_out	1’d1	yes
3	lp_uart_ctsn_pad_in	1	no	lp_uart_rtsn_pad_out	1’d1	no
4	lp_uart_dsrn_pad_in	1	no	lp_uart_dtrn_pad_out	1’d1	no
5	lp_spi_ck_pad_in	0	no	lp_spi_ck_pad_out	lp_spi_ck_pad_oe	no
6	lp_spi_cs_pad_in	0	no	lp_spi_cs_pad_out	lp_spi_cs_pad_oe	no
7	lp_spi_d_pad_in	0	no	lp_spi_d_pad_out	lp_spi_d_pad_oe	no
8	lp_spi_q_pad_in	0	no	lp_spi_q_pad_out	lp_spi_q_pad_oe	no
9	lp_i2s_i_bck_pad_in	0	no	lp_i2s_i_bck_pad_out	1’d1	no
10	lp_i2s_i_sd_pad_in	0	no	lp_i2s_o_sd_pad_out	1’d1	no
11	lp_i2s_i_ws_pad_in	0	no	lp_i2s_i_ws_pad_out	1’d1	no
12	lp_i2s_o_bck_pad_in	0	no	lp_i2s_o_bck_pad_out	1’d1	no
13	lp_i2s_o_ws_pad_in	0	no	lp_i2s_o_ws_pad_out	1’d1	no
14	-	-	-	-	-	-
15	-	-	-	-	-	-
16	-	-	-	-	-	-
17	-	-	-	-	-	-
18	-	-	-	-	-	-
19	-	-	-	-	-	-
20	-	-	-	-	-	-
21	-	-	-	-	-	-
22	-	-	-	-	-	-
23	-	-	-	-	-	-

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Signal No.	Input Signal	Default value	Direct Input via LP IO MUX	Output Signal	Output enable signal when LP_GPIO_FUNCn_OE_SEL = 0	Direct Output via LP IO MUX
24	-	-	-	-	-	-
25	-	-	-	-	-	-
26	-	-	-	-	-	-
27	-	-	-	-	-	-
28	-	-	-	-	-	-
29	-	-	-	-	-	-
30	-	-	-	-	-	-
31	-	-	-	-	-	-

9.14 HP IO MUX Functions List

Table 9.14-1 shows the HP IO MUX functions and default states of each HP GPIO pin.

Table 9.14-1. HP IO MUX Pin Functions

Name	Function 0	Function 1	Function 2	Function 3	DRV	Reset	Notes
GPIO0	GPIO0	GPIO0	-	-	2	0	R
GPIO1	GPIO1	GPIO1	-	-	2	0	R
GPIO2	MTCK	GPIO2	-	-	2	1*	R
GPIO3	MTDI	GPIO3	-	-	2	1	R
GPIO4	MTMS	GPIO4	-	-	2	1	R
GPIO5	MTDO	GPIO5	-	-	2	0	R
GPIO6	GPIO6	GPIO6	-	SPI2_HOLD_PAD	2	0	R
GPIO7	GPIO7	GPIO7	-	SPI2_CS_PAD	2	0	R
GPIO8	GPIO8	GPIO8	UART0_RTS_PAD	SPI2_D_PAD	2	0	R

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Table 9.14-1 – cont'd from previous page

Name	Function 0	Function 1	Function 2	Function 3	DRV	Reset	Notes
GPIO9	GPIO9	GPIO9	UART0_CTS_PAD	SPI2_CK_PAD	2	0	R
GPIO10	GPIO10	GPIO10	UART1_TXD_PAD	SPI2_Q_PAD	2	0	R
GPIO11	GPIO11	GPIO11	UART1_RXD_PAD	SPI2_WP_PAD	2	0	R
GPIO12	GPIO12	GPIO12	UART1_RTS_PAD	-	2	0	R
GPIO13	GPIO13	GPIO13	UART1_CTS_PAD	-	2	0	R
GPIO14	GPIO14	GPIO14	—	-	2	0	R
GPIO15	GPIO15	GPIO15	—	-	2	0	R
GPIO16	GPIO16	GPIO16	—	-	2	0	R
GPIO17	GPIO17	GPIO17	—	-	2	0	R
GPIO18	GPIO18	GPIO18	—	-	2	0	R
GPIO19	GPIO19	GPIO19	—	-	2	0	R
GPIO20	GPIO20	GPIO20	—	-	2	0	R
GPIO21	GPIO21	GPIO21	—	-	2	0	R
GPIO22	GPIO22	GPIO22	—	-	2	0	R
GPIO23	GPIO23	GPIO23	—	REF_50M_CLK_PAD	2	0	R
GPIO24	GPIO24	GPIO24	—	-	3	0	R, USB
GPIO25	GPIO25	GPIO25	—	-	3	3*	R, USB
GPIO26	GPIO26	GPIO26	—	-	2	0	R, USB
GPIO27	GPIO27	GPIO27	—	-	2	0	R, USB
GPIO28	GPIO28	GPIO28	SPI2_CS_PAD	GMAC_PHY_RXDV_PAD	2	0	—
GPIO29	GPIO29	GPIO29	SPI2_D_PAD	GMAC_PHY_RXD0_PAD	2	0	—
GPIO30	GPIO30	GPIO30	SPI2_CK_PAD	GMAC_PHY_RXD1_PAD	2	0	—
GPIO31	GPIO31	GPIO31	SPI2_Q_PAD	GMAC_PHY_RXER_PAD	2	0	—
GPIO32	GPIO32	GPIO32	SPI2_HOLD_PAD	GMAC_RMII_CLK_PAD	2	0	—
GPIO33	GPIO33	GPIO33	SPI2_WP_PAD	GMAC_PHY_TXEN_PAD	2	0	—
GPIO34	GPIO34	GPIO34	SPI2_IO4_PAD	GMAC_PHY_TXD0_PAD	2	0	—
GPIO35	GPIO35	GPIO35	SPI2_IO5_PAD	GMAC_PHY_TXD1_PAD	2	0	—

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Table 9.14-1 – cont'd from previous page

Name	Function 0	Function 1	Function 2	Function 3	DRV	Reset	Notes
GPIO36	GPIO36	GPIO36	SPI2_IO6_PAD	GMAC_PHY_TXER_PAD	2	0	—
GPIO37	UART0_TXD_PAD	GPIO37	SPI2_IO7_PAD	-	2	5	—
GPIO38	UART0_RXD_PAD	GPIO38	SPI2_DQS_PAD	-	2	0	—
GPIO39	SD1_CDATA0_PAD	GPIO39	—	REF_50M_CLK_PAD	2	0	—
GPIO40	SD1_CDATA1_PAD	GPIO40	—	GMAC_PHY_TXEN_PAD	2	0	—
GPIO41	SD1_CDATA2_PAD	GPIO41	—	GMAC_PHY_TXD0_PAD	2	0	—
GPIO42	SD1_CDATA3_PAD	GPIO42	—	GMAC_PHY_TXD1_PAD	2	0	—
GPIO43	SD1_CCLK_PAD	GPIO43	—	GMAC_PHY_TXER_PAD	2	0	—
GPIO44	SD1_CCMD_PAD	GPIO44	—	GMAC_RMII_CLK_PAD	2	0	—
GPIO45	SD1_CDATA4_PAD	GPIO45	—	GMAC_PHY_RXDV_PAD	2	0	—
GPIO46	SD1_CDATA5_PAD	GPIO46	—	GMAC_PHY_RXD0_PAD	2	0	—
GPIO47	SD1_CDATA6_PAD	GPIO47	—	GMAC_PHY_RXD1_PAD	2	0	—
GPIO48	SD1_CDATA7_PAD	GPIO48	—	GMAC_PHY_RXER_PAD	2	0	—
GPIO49	GPIO49	GPIO49	—	GMAC_PHY_TXEN_PAD	2	0	R
GPIO50	GPIO50	GPIO50	—	GMAC_RMII_CLK_PAD	2	0	R
GPIO51	GPIO51	GPIO51	—	GMAC_PHY_RXDV_PAD	2	0	R
GPIO52	GPIO52	GPIO52	—	GMAC_PHY_RXD0_PAD	2	0	R
GPIO53	GPIO53	GPIO53	—	GMAC_PHY_RXD1_PAD	2	0	R
GPIO54	GPIO54	GPIO54	—	GMAC_PHY_RXER_PAD	2	0	R

Drive Strength

"DRV" column shows the drive strength of each pin after reset:

- **0** - Drive current = ~5 mA
- **1** - Drive current = ~10 mA
- **2** - Drive current = ~20 mA
- **3** - Drive current = ~40 mA

Reset

The default configuration of each pin after reset:

- **0** - input disabled, in high impedance state (IE = 0)
- **1** - input enabled, in high impedance state (IE = 1)
- **1*** - When the value of eFuse bit EFUSE_DIS_PAD_JTAG is
  0 (default), input enabled, pull-up resistor enabled (IE = 1, WPU = 1)
  1, input disabled, in high impedance state (IE = 1)
- **3*** - input disabled, pull-up resistor enabled (IE = 0, USB_PU = 1). See details in Notes.
- **5** - input disabled (IE = 0). Output is controlled by the peripheral of Function 0, and the default output is 1.

We recommend pulling high or low GPIO pins in high impedance state to avoid unnecessary power consumption. You may add pull-up and pull-down resistors in your PCB design, or enable internal pull-up and pull-down resistors during software initialization.

Notes

- **R** - These pins have analog functions.
- **USB** - The pull-up value of a USB pin is controlled by the pin's pull-up value together with the USB pull-up value. If any of the two pull-up values is 1, the pin's pull-up resistor will be enabled.

Notice:
In HP IO MUX, unused pins must be configured to GPIO function.

9.15 LP IO MUX Functions List

Table 9.15-1 shows the LP IO MUX functions of each GPIO pin. GPIO pins are default controlled by HP IO MUX, so Table 9.15-1 only show functions of LP IO MUX.

Table 9.15-1. LP IO MUX Pin Functions

Name	Function 0	Function 1
GPIO0	LP_GPIO0	LP_GPIO0
GPIO1	LP_GPIO1	LP_GPIO1
GPIO2	LP_GPIO2	LP_GPIO2

Name	Function 0	Function 1
GPIO3	LP_GPIO3	LP_GPIO3
GPIO4	LP_GPIO4	LP_GPIO4
GPIO5	LP_GPIO5	LP_GPIO5
GPIO6	LP_GPIO6	LP_GPIO6
GPIO7	LP_GPIO7	LP_GPIO7
GPIO8	LP_GPIO8	LP_GPIO8
GPIO9	LP_GPIO9	LP_GPIO9
GPIO10	LP_GPIO10	LP_GPIO10
GPIO11	LP_GPIO11	LP_GPIO11
GPIO12	LP_GPIO12	LP_GPIO12
GPIO13	LP_GPIO13	LP_GPIO13
GPIO14	LP_UART_TXD_PAD	LP_GPIO14
GPIO15	LP_UART_RXD_PAD	LP_GPIO15

Notice:

In LP IO MUX, unused pins must be configured to LP_GPIO function.

9.16 GPIO Pin Analog Functions List

Table 9.16-1 shows the GPIO pins and their correspond analog functions.

Table 9.16-1. GPIO Pin Analog Functions

Name	Analog Function 0	Analog Function 1
GPIO0	XTAL_32K_N	—
GPIO1	XTAL_32K_P	—
GPIO2	TOUCH_CHANNEL0	—
GPIO3	TOUCH_CHANNEL1	—
GPIO4	TOUCH_CHANNEL2	—
GPIO5	TOUCH_CHANNEL3	—
GPIO6	TOUCH_CHANNEL4	—
GPIO7	TOUCH_CHANNEL5	—
GPIO8	TOUCH_CHANNEL6	—
GPIO9	TOUCH_CHANNEL7	—
GPIO10	TOUCH_CHANNEL8	—
GPIO11	TOUCH_CHANNEL9	—
GPIO12	TOUCH_CHANNEL10	—
GPIO13	TOUCH_CHANNEL11	—
GPIO14	TOUCH_CHANNEL12	—
GPIO15	TOUCH_CHANNEL13	—
GPIO16	ADC1_CHANNEL0	—
GPIO17	ADC1_CHANNEL1	—

Name	Analog Function 0	Analog Function 1
GPIO18	ADC1_CHANNEL2	—
GPIO19	ADC1_CHANNEL3	—
GPIO20	ADC1_CHANNEL4	—
GPIO21	ADC1_CHANNEL5	—
GPIO22	ADC1_CHANNEL6	—
GPIO23	ADC1_CHANNEL7	—
GPIO24	USB1P1_N0	—
GPIO25	USB1P1_P0	—
GPIO26	USB1P1_N1	—
GPIO27	USB1P1_P1	—
GPIO49	ADC2_CHANNEL0	—
GPIO50	ADC2_CHANNEL1	—
GPIO51	ADC2_CHANNEL2	ANA_COMP0
GPIO52	ADC2_CHANNEL3	ANA_COMP0
GPIO53	ADC2_CHANNEL4	ANA_COMP1
GPIO54	ADC2_CHANNEL5	ANA_COMP1

9.17 Event Task Matrix Function

In ESP32-P4, GPIO supports ETM function, that is, the ETM task of GPIO can be triggered by the ETM event of any peripheral, or the ETM task of any peripheral can be triggered by the ETM event of GPIO. For more details about ETM, please refer to Chapter 13 *Event Task Matrix (ETM)*. Only ETM tasks and ETM events related to GPIO are introduced here.

The GPIO ETM provides eight task channels *x* (0 ~ 7). The ETM tasks that each task channel can receive are:

- GPIO_TASK_CH*x*_SET: GPIO goes high when triggered.
- GPIO_TASK_CH*x*_CLEAR: GPIO goes low when triggered.
- GPIO_TASK_CH*x*_TOGGLE: GPIO toggles level when triggered.

Below is an example to configure task channel *x* to control GPIO*y*:

- Configure IO_MUX_GPIO*y*_MCU_SEL to 1, to select Function 1 listed in Table 9.14-1.
- Configure GPIO_ENABLE_REG[*y*] to 1.
- Configure GPIO_EXT_ETM_TASK_GPIO*y*_SEL to *x*.
- Set GPIO_EXT_ETM_TASK_GPIO*y*_EN, to enable ETM task channel *x* to control GPIO*y*.

Note:

- One task channel can be selected by one or more GPIOs.
- When two or three of the signals GPIO_TASK_CH*x*_SET, GPIO_TASK_CH*x*_CLEAR, and GPIO_TASK_CH*x*_TOGGLE of the task channel *x* selected by GPIO*y* are valid at the same time, then GPIO_TASK_CH*x*_SET has the highest priority, GPIO_TASK_CH*x*_CLEAR takes the second higher priority, and GPIO_TASK_CH*x*_TOGGLE has the

lowest priority.

- When GPIO*y* is controlled by ETM task channel, the values of GPIO_OUT_REG, GPIO_FUNC*n*_OUT_INV_SEL, and GPIO_FUNC*n*_OUT_SEL may be modified by the hardware. For such reason, it's recommended to reconfigure these registers when the GPIO is free from the control of ETM task channel.

GPIO has eight event channels, and the ETM events that each event channel can generate are:

- GPIO_EVT_CH*x*_RISE_EDGE: Indicates that the output signal of the corresponding GPIO Filter (see Figure 9.3-1) has a rising edge.
- GPIO_EVT_CH*x*_FALL_EDGE: Indicates that the output signal of the corresponding GPIO Filter (see Figure 9.3-1) has a falling edge.
- GPIO_EVT_CH*x*_ANY_EDGE: Indicates that the output signal of the corresponding GPIO Filter (see Figure 9.3-1) is reversed.

The specific configuration of the event channel is as follows:

- Set GPIO_EXT_ETM_CH*x*_EVENT_EN to enable event channel *x* (0 ~ 7).
- Configure GPIO_EXT_ETM_CH*x*_EVENT_SEL to *y* (0 ~ 54), i.e., select one from the 55 GPIOs.

Note:
One GPIO can be selected by one or more event channels.

In specific applications, GPIO ETM events can be used to trigger GPIO ETM tasks. For example, event channel 0 selects GPIO0, GPIO1 selects task channel 0, and the GPIO_EVT_CH*0*_RISE_EDGE event is used to trigger the GPIO_TASK_CH*0*_TOGGLE task. When a square wave signal is input to the chip through GPIO0, the chip outputs a square wave signal with a frequency divided by 2 through GPIO1.

9.18 Interrupts

ESP32-P4's HP IO MUX and HP GPIO matrix can generate the following interrupt signals that will be sent to the Interrupt Matrix.

- GPIO_INTR0
- GPIO_INTR1
- GPIO_INTR2
- GPIO_INTR3
- GPIO_PAD_COMP_INTR

There are several internal interrupt sources from HP IO MUX and HP GPIO matrix that can generate the above interrupt signals. The interrupt sources from HP IO MUX and HP GPIO matrix are listed with their trigger conditions and the resulted interrupt signals in Table 9.18-1. For more detailed information about GPIO_PAD_COMP_INTR, please refer to Chapter 63 *Analog Voltage Comparator*.

Table 9.18-1. HP IO MUX and HP GPIO Matrix's Internal Interrupt Sources

Internal Interrupt Source	Trigger Condition	Interrupt Signal

GPIO_INTR0	GPIO_STATUS_INTERRUPT[n] & GPIO_PINn_INT_ENA[0] (n: 0 ~ 31)	GPIO_INTR0
GPIO_INTR0	GPIO_STATUS1_INTERRUPT[n] & GPIO_PINn+32_INT_ENA[0] (n: 0 ~ 22)	GPIO_INTR0
GPIO_INTR1	GPIO_STATUS_INTERRUPT[n] & GPIO_PINn_INT_ENA[1] (n: 0 ~ 31)	GPIO_INTR1
GPIO_INTR1	GPIO_STATUS1_INTERRUPT[n] & GPIO_PINn+32_INT_ENA[1] (n: 0 ~ 22)	GPIO_INTR1
GPIO_INTR2	GPIO_STATUS_INTERRUPT[n] & GPIO_PINn_INT_ENA[3] (n: 0 ~ 31)	GPIO_INTR2
GPIO_INTR2	GPIO_STATUS1_INTERRUPT[n] & GPIO_PINn+32_INT_ENA[3] (n: 0 ~ 22)	GPIO_INTR2
GPIO_INTR3	GPIO_STATUS_INTERRUPT[n] & GPIO_PINn_INT_ENA[4] (n: 0 ~ 31)	GPIO_INTR3
GPIO_INTR3	GPIO_STATUS1_INTERRUPT[n] & GPIO_PINn+32_INT_ENA[4] (n: 0 ~ 22)	GPIO_INTR3

ESP32-P4's LP IO MUX and GPIO matrix can generate the following interrupt signals that will be sent to the Interrupt Matrix.

- LP_GPIO_INTR0

The interrupt source from LP IO MUX and LP GPIO matrix are listed with their trigger conditions and the resulted interrupt signals in Table 9.18-2.

Table 9.18-2. LP IO MUX and LP GPIO Matrix's Internal Interrupt Sources

Internal Interrupt Source	Trigger Condition	Interrupt Signal
LP_GPIO_INTR	LP_GPIO_STATUS_DATA[n] (n: 0 ~ 15)	LP_GPIO_INTR

9.19 Register Summary

9.19.1 HP GPIO Matrix Register Summary

The addresses in this section are relative to HP GPIO matrix base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

The abbreviations given in Column **Access** are explained in Section *Access Types for Registers*.

Name	Description	Address	Access
Configuration Registers			
GPIO_OUT_REG	GPIO0 ~ GPIO31 output register	0x0004	R/W/SC/WTC
GPIO_OUT_W1TS_REG	GPIO0 ~ GPIO31 output set register	0x0008	WT
GPIO_OUT_W1TC_REG	GPIO0 ~ GPIO31 output clear register	0x000C	WT
GPIO_OUT1_REG	GPIO32 ~ GPIO56 output register	0x0010	R/W/SC/WTC
GPIO_OUT1_W1TS_REG	GPIO32 ~ GPIO56 output set register	0x0014	WT

Name	Description	Address	Access
GPIO_OUT1_W1TC_REG	GPIO32 ~ GPIO56 output clear register	0x0018	WT
GPIO_ENABLE_REG	GPIO0 ~ GPIO31 output enable register	0x0020	R/W/WTC
GPIO_ENABLE_W1TS_REG	GPIO0 ~ GPIO31 output enable set register	0x0024	WT
GPIO_ENABLE_W1TC_REG	GPIO0 ~ GPIO31 output enable clear register	0x0028	WT
GPIO_ENABLE1_REG	GPIO32 ~ GPIO56 output enable register	0x002C	R/W/WTC
GPIO_ENABLE1_W1TS_REG	GPIO32 ~ GPIO56 output enable set register	0x0030	WT
GPIO_ENABLE1_W1TC_REG	GPIO32 ~ GPIO56 output enable clear register	0x0034	WT
GPIO_STRAP_REG	Strapping pin register	0x0038	RO
GPIO_IN_REG	GPIO0 ~ GPIO31 input register	0x003C	RO
GPIO_IN1_REG	GPIO32 ~ GPIO56 input register	0x0040	RO
GPIO_STATUS_REG	GPIO0 ~ GPIO31 interrupt status register	0x0044	R/W/WTC
GPIO_STATUS_W1TS_REG	GPIO0 ~ GPIO31 interrupt status set register	0x0048	WT
GPIO_STATUS_W1TC_REG	GPIO0 ~ GPIO31 interrupt status clear register	0x004C	WT
GPIO_STATUS1_REG	GPIO32 ~ GPIO56 interrupt status register	0x0050	R/W/WTC
GPIO_STATUS1_W1TS_REG	GPIO32 ~ GPIO56 interrupt status set register	0x0054	WT
GPIO_STATUS1_W1TC_REG	GPIO32 ~ GPIO56 interrupt status clear register	0x0058	WT
GPIO_INTR_0_REG	GPIO0 ~ GPIO31 interrupt 0 status register	0x005C	RO
GPIO_INTR1_0_REG	GPIO32 ~ GPIO56 interrupt 0 status register	0x0060	RO
GPIO_INTR_1_REG	GPIO0 ~ GPIO31 interrupt 1 status register	0x0064	RO
GPIO_INTR1_1_REG	GPIO32 ~ GPIO56 interrupt 1 status register	0x0068	RO
GPIO_STATUS_NEXT_REG	GPIO0 ~ GPIO31 interrupt source register	0x006C	RO
GPIO_STATUS_NEXT1_REG	GPIO32 ~ GPIO56 interrupt source register	0x0070	RO
GPIO_PIN0_REG	GPIO pin0 configuration register	0x0074	R/W
GPIO_PIN1_REG	GPIO pin1 configuration register	0x0078	R/W
GPIO_PIN2_REG	GPIO pin2 configuration register	0x007C	R/W
GPIO_PIN3_REG	GPIO pin3 configuration register	0x0080	R/W
GPIO_PIN4_REG	GPIO pin4 configuration register	0x0084	R/W
...	...	...	...
GPIO_PIN54_REG	GPIO pin54 configuration register	0x014C	R/W
GPIO_PIN55_REG	GPIO pin55 configuration register	0x0150	R/W
GPIO_PIN56_REG	GPIO pin56 configuration register	0x0154	R/W
GPIO_FUNC1_IN_SEL_CFG_REG	Input signal 1 configuration register	0x015C	R/W
GPIO_FUNC2_IN_SEL_CFG_REG	Input signal 2 configuration register	0x0160	R/W
GPIO_FUNC3_IN_SEL_CFG_REG	Input signal 3 configuration register	0x0164	R/W
...	...	...	...
GPIO_FUNC253_IN_SEL_CFG_REG	Input signal 253 configuration register	0x054C	R/W
GPIO_FUNC254_IN_SEL_CFG_REG	Input signal 254 configuration register	0x0550	R/W
GPIO_FUNC255_IN_SEL_CFG_REG	Input signal 255 configuration register	0x0554	R/W
GPIO_FUNC0_OUT_SEL_CFG_REG	GPIO0 output configuration register	0x0558	varies
GPIO_FUNC1_OUT_SEL_CFG_REG	GPIO1 output configuration register	0x055C	varies
GPIO_FUNC2_OUT_SEL_CFG_REG	GPIO2 output configuration register	0x0560	varies
GPIO_FUNC3_OUT_SEL_CFG_REG	GPIO3 output configuration register	0x0564	varies
...	...	...	...

Name	Description	Address	Access
GPIO_FUNC53_OUT_SEL_CFG_REG	GPIO53 output configuration register	0x062C	varies
GPIO_FUNC54_OUT_SEL_CFG_REG	GPIO54 output configuration register	0x0630	varies
GPIO_FUNC55_OUT_SEL_CFG_REG	GPIO55 output configuration register	0x0634	varies
GPIO_FUNC56_OUT_SEL_CFG_REG	GPIO56 output configuration register	0x0638	varies
GPIO_INTR_2_REG	GPIO0 ~ GPIO31 interrupt 2 status register	0x063C	RO
GPIO_INTR1_2_REG	GPIO32 ~ GPIO56 interrupt 2 status register	0x0640	RO
GPIO_INTR_3_REG	GPIO0 ~ GPIO31 interrupt 3 status register	0x0644	RO
GPIO_INTR1_3_REG	GPIO32 ~ GPIO56 interrupt 3 status register	0x0648	RO
GPIO_CLOCK_GATE_REG	GPIO clock gate register	0x064C	R/W
GPIO_ZERO_DET0_FILTER_CNT_REG	GPIO analog comparator zero detect filter count	0x0710	R/W
GPIO_ZERO_DET1_FILTER_CNT_REG	GPIO analog comparator zero detect filter count	0x0714	R/W
GPIO Interrupt Registers			
GPIO_INT_RAW_REG	Analog comparator interrupt raw register	0x0700	R/WTC/SS
GPIO_INT_ST_REG	Analog comparator interrupt status register	0x0704	RO
GPIO_INT_ENA_REG	Analog comparator interrupt enable register	0x0708	R/W
GPIO_INT_CLR_REG	Analog comparator interrupt clear register	0x070C	WT
Version Register			
GPIO_DATE_REG	Version control register	0x07FC	R/W

9.19.2 HP IO MUX Register Summary

The addresses in this section are relative to the HP IO MUX base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

The abbreviations given in Column **Access** are explained in Section *Access Types for Registers*.

Name	Description	Address	Access
IO MUX Configuration Registers			
IO_MUX_GPIO0_REG	IO MUX configuration register for GPIO0	0x0004	R/W
IO_MUX_GPIO1_REG	IO MUX configuration register for GPIO1	0x0008	R/W
IO_MUX_GPIO2_REG	IO MUX configuration register for GPIO2	0x000C	R/W
IO_MUX_GPIO3_REG	IO MUX configuration register for GPIO3	0x0010	R/W
IO_MUX_GPIO4_REG	IO MUX configuration register for GPIO4	0x0014	R/W
IO_MUX_GPIO5_REG	IO MUX configuration register for GPIO5	0x0018	R/W
IO_MUX_GPIO6_REG	IO MUX configuration register for GPIO6	0x001C	R/W
IO_MUX_GPIO7_REG	IO MUX configuration register for GPIO7	0x0020	R/W
IO_MUX_GPIO8_REG	IO MUX configuration register for GPIO8	0x0024	R/W
IO_MUX_GPIO9_REG	IO MUX configuration register for GPIO9	0x0028	R/W
IO_MUX_GPIO10_REG	IO MUX configuration register for GPIO10	0x002C	R/W
IO_MUX_GPIO11_REG	IO MUX configuration register for GPIO11	0x0030	R/W
IO_MUX_GPIO12_REG	IO MUX configuration register for GPIO12	0x0034	R/W
IO_MUX_GPIO13_REG	IO MUX configuration register for GPIO13	0x0038	R/W
IO_MUX_GPIO14_REG	IO MUX configuration register for GPIO14	0x003C	R/W
IO_MUX_GPIO15_REG	IO MUX configuration register for GPIO15	0x0040	R/W

Name	Description	Address	Access
IO_MUX_GPIO16_REG	IO MUX configuration register for GPIO16	0x0044	R/W
IO_MUX_GPIO17_REG	IO MUX configuration register for GPIO17	0x0048	R/W
IO_MUX_GPIO18_REG	IO MUX configuration register for GPIO18	0x004C	R/W
IO_MUX_GPIO19_REG	IO MUX configuration register for GPIO19	0x0050	R/W
IO_MUX_GPIO20_REG	IO MUX configuration register for GPIO20	0x0054	R/W
IO_MUX_GPIO21_REG	IO MUX configuration register for GPIO21	0x0058	R/W
IO_MUX_GPIO22_REG	IO MUX configuration register for GPIO22	0x005C	R/W
IO_MUX_GPIO23_REG	IO MUX configuration register for GPIO23	0x0060	R/W
IO_MUX_GPIO24_REG	IO MUX configuration register for GPIO24	0x0064	R/W
IO_MUX_GPIO25_REG	IO MUX configuration register for GPIO25	0x0068	R/W
IO_MUX_GPIO26_REG	IO MUX configuration register for GPIO26	0x006C	R/W
IO_MUX_GPIO27_REG	IO MUX configuration register for GPIO27	0x0070	R/W
IO_MUX_GPIO28_REG	IO MUX configuration register for GPIO28	0x0074	R/W
IO_MUX_GPIO29_REG	IO MUX configuration register for GPIO29	0x0078	R/W
IO_MUX_GPIO30_REG	IO MUX configuration register for GPIO30	0x007C	R/W
IO_MUX_GPIO31_REG	IO MUX configuration register for GPIO31	0x0080	R/W
IO_MUX_GPIO32_REG	IO MUX configuration register for GPIO32	0x0084	R/W
IO_MUX_GPIO33_REG	IO MUX configuration register for GPIO33	0x0088	R/W
IO_MUX_GPIO34_REG	IO MUX configuration register for GPIO34	0x008C	R/W
IO_MUX_GPIO35_REG	IO MUX configuration register for GPIO35	0x0090	R/W
IO_MUX_GPIO36_REG	IO MUX configuration register for GPIO36	0x0094	R/W
IO_MUX_GPIO37_REG	IO MUX configuration register for GPIO37	0x0098	R/W
IO_MUX_GPIO38_REG	IO MUX configuration register for GPIO38	0x009C	R/W
IO_MUX_GPIO39_REG	IO MUX configuration register for GPIO39	0x00A0	R/W
IO_MUX_GPIO40_REG	IO MUX configuration register for GPIO40	0x00A4	R/W
IO_MUX_GPIO41_REG	IO MUX configuration register for GPIO41	0x00A8	R/W
IO_MUX_GPIO42_REG	IO MUX configuration register for GPIO42	0x00AC	R/W
IO_MUX_GPIO43_REG	IO MUX configuration register for GPIO43	0x00B0	R/W
IO_MUX_GPIO44_REG	IO MUX configuration register for GPIO44	0x00B4	R/W
IO_MUX_GPIO45_REG	IO MUX configuration register for GPIO45	0x00B8	R/W
IO_MUX_GPIO46_REG	IO MUX configuration register for GPIO46	0x00BC	R/W
IO_MUX_GPIO47_REG	IO MUX configuration register for GPIO47	0x00C0	R/W
IO_MUX_GPIO48_REG	IO MUX configuration register for GPIO48	0x00C4	R/W
IO_MUX_GPIO49_REG	IO MUX configuration register for GPIO49	0x00C8	R/W
IO_MUX_GPIO50_REG	IO MUX configuration register for GPIO50	0x00CC	R/W
IO_MUX_GPIO51_REG	IO MUX configuration register for GPIO51	0x00D0	R/W
IO_MUX_GPIO52_REG	IO MUX configuration register for GPIO52	0x00D4	R/W
IO_MUX_GPIO53_REG	IO MUX configuration register for GPIO53	0x00D8	R/W
IO_MUX_GPIO54_REG	IO MUX configuration register for GPIO54	0x00DC	R/W
Version Register			
IO_MUX_DATE_REG	Version control register	0x0104	R/W

9.19.3 GPIO EXT Register Summary

GPIO EXT registers consist of SDM registers, Glitch Filter registers, and ETM registers.

The addresses in this section are relative to (HP GPIO matrix base address + 0x0F00). GPIO base address is provided in Table 7.3-2 in Chapter 7 *System and Memory*.

The abbreviations given in Column **Access** are explained in Section *Access Types for Registers*.

Name	Description	Address	Access
SDM Configuration Registers			
GPIO_EXT_SIGMADELTA0_REG	SDM channel 0 duty cycle configuration register	0x0000	R/W
GPIO_EXT_SIGMADELTA1_REG	SDM channel 1 duty cycle configuration register	0x0004	R/W
GPIO_EXT_SIGMADELTA2_REG	SDM channel 2 duty cycle configuration register	0x0008	R/W
GPIO_EXT_SIGMADELTA3_REG	SDM channel 3 duty cycle configuration register	0x000C	R/W
GPIO_EXT_SIGMADELTA4_REG	SDM channel 4 duty cycle configuration register	0x0010	R/W
GPIO_EXT_SIGMADELTA5_REG	SDM channel 5 duty cycle configuration register	0x0014	R/W
GPIO_EXT_SIGMADELTA6_REG	SDM channel 6 duty cycle configuration register	0x0018	R/W
GPIO_EXT_SIGMADELTA7_REG	SDM channel 7 duty cycle configuration register	0x001C	R/W
GPIO_EXT_SIGMADELTA_MISC_REG	MISC register	0x0024	R/W
Glitch Filter Configuration Registers			
GPIO_EXT_GLITCH_FILTER_CH0_REG	Glitch Filter configuration register for channel 0	0x0030	R/W
GPIO_EXT_GLITCH_FILTER_CH1_REG	Glitch Filter configuration register for channel 1	0x0034	R/W
GPIO_EXT_GLITCH_FILTER_CH2_REG	Glitch Filter configuration register for channel 2	0x0038	R/W
GPIO_EXT_GLITCH_FILTER_CH3_REG	Glitch Filter configuration register for channel 3	0x003C	R/W
GPIO_EXT_GLITCH_FILTER_CH4_REG	Glitch Filter configuration register for channel 4	0x0040	R/W
GPIO_EXT_GLITCH_FILTER_CH5_REG	Glitch Filter configuration register for channel 5	0x0044	R/W
GPIO_EXT_GLITCH_FILTER_CH6_REG	Glitch Filter configuration register for channel 6	0x0048	R/W
GPIO_EXT_GLITCH_FILTER_CH7_REG	Glitch Filter configuration register for channel 7	0x004C	R/W
ETM Configuration Registers			
GPIO_EXT_ETM_EVENT_CH0_CFG_REG	ETM configuration register for channel 0	0x0060	R/W

Name	Description	Address	Access
GPIO_EXT_ETM_EVENT_CH1_CFG_REG	ETM configuration register for channel 1	0x0064	R/W
GPIO_EXT_ETM_EVENT_CH2_CFG_REG	ETM configuration register for channel 2	0x0068	R/W
GPIO_EXT_ETM_EVENT_CH3_CFG_REG	ETM configuration register for channel 3	0x006C	R/W
GPIO_EXT_ETM_EVENT_CH4_CFG_REG	ETM configuration register for channel 4	0x0070	R/W
GPIO_EXT_ETM_EVENT_CH5_CFG_REG	ETM configuration register for channel 5	0x0074	R/W
GPIO_EXT_ETM_EVENT_CH6_CFG_REG	ETM configuration register for channel 6	0x0078	R/W
GPIO_EXT_ETM_EVENT_CH7_CFG_REG	ETM configuration register for channel 7	0x007C	R/W
GPIO_EXT_ETM_TASK_P0_CFG_REG	GPIO selection register 0 for ETM	0x00A0	R/W
GPIO_EXT_ETM_TASK_P1_CFG_REG	GPIO selection register 1 for ETM	0x00A4	R/W
GPIO_EXT_ETM_TASK_P2_CFG_REG	GPIO selection register 2 for ETM	0x00A8	R/W
GPIO_EXT_ETM_TASK_P3_CFG_REG	GPIO selection register 3 for ETM	0x00AC	R/W
GPIO_EXT_ETM_TASK_P4_CFG_REG	GPIO selection register 4 for ETM	0x00B0	R/W
GPIO_EXT_ETM_TASK_P5_CFG_REG	GPIO selection register 5 for ETM	0x00B4	R/W
GPIO_EXT_ETM_TASK_P6_CFG_REG	GPIO selection register 6 for ETM	0x00B8	R/W
GPIO_EXT_ETM_TASK_P7_CFG_REG	GPIO selection register 7 for ETM	0x00BC	R/W
GPIO_EXT_ETM_TASK_P8_CFG_REG	GPIO selection register 8 for ETM	0x00C0	R/W
GPIO_EXT_ETM_TASK_P9_CFG_REG	GPIO selection register 9 for ETM	0x00C4	R/W
GPIO_EXT_ETM_TASK_P10_CFG_REG	GPIO selection register 10 for ETM	0x00C8	R/W
GPIO_EXT_ETM_TASK_P11_CFG_REG	GPIO selection register 11 for ETM	0x00CC	R/W
GPIO_EXT_ETM_TASK_P12_CFG_REG	GPIO selection register 12 for ETM	0x00D0	R/W
GPIO_EXT_ETM_TASK_P13_CFG_REG	GPIO selection register 13 for ETM	0x00D4	R/W
Version Register			
GPIO_EXT_VERSION_REG	Version control register	0x00FC	R/W

9.19.4 LP GPIO Matrix Register Summary

The addresses in this section are relative to LP GPIO matrix base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

The abbreviations given in Column **Access** are explained in Section *Access Types for Registers*.

Name	Description	Address	Access
Clock Enable Register			
LP_GPIO_CLK_EN_REG	LP GPIO clock gate register	0x0000	R/W
Configuration Registers			
LP_GPIO_OUT_REG	LP GPIO output register	0x0008	R/W/WTC
LP_GPIO_OUT_W1TS_REG	LP GPIO output set register	0x000C	WT
LP_GPIO_OUT_W1TC_REG	LP GPIO output clear register	0x0010	WT
LP_GPIO_ENABLE_REG	LP GPIO output enable register	0x0014	R/W/WTC
LP_GPIO_ENABLE_W1TS_REG	LP GPIO output enable set register	0x0018	WT
LP_GPIO_ENABLE_W1TC_REG	LP GPIO output enable clear register	0x001C	WT
LP_GPIO_STATUS_REG	LP GPIO interrupt status register	0x0020	R/W/WTC
LP_GPIO_STATUS_W1TS_REG	LP GPIO interrupt status set register	0x0024	WT
LP_GPIO_STATUS_W1TC_REG	LP GPIO interrupt status clear register	0x0028	WT

Name	Description	Address	Access
LP_GPIO_STATUS_NEXT_REG	LP GPIO interrupt source register	0x002C	RO
LP_GPIO_IN_REG	LP GPIO input register	0x0030	RO
LP_GPIO_PIN0_REG	LP GPIO0 configuration register	0x0034	varies
LP_GPIO_PIN1_REG	LP GPIO1 configuration register	0x0038	varies
LP_GPIO_PIN2_REG	LP GPIO2 configuration register	0x003C	varies
LP_GPIO_PIN3_REG	LP GPIO3 configuration register	0x0040	varies
LP_GPIO_PIN4_REG	LP GPIO4 configuration register	0x0044	varies
LP_GPIO_PIN5_REG	LP GPIO5 configuration register	0x0048	varies
LP_GPIO_PIN6_REG	LP GPIO6 configuration register	0x004C	varies
LP_GPIO_PIN7_REG	LP GPIO7 configuration register	0x0050	varies
LP_GPIO_PIN8_REG	LP GPIO8 configuration register	0x0054	varies
LP_GPIO_PIN9_REG	LP GPIO9 configuration register	0x0058	varies
LP_GPIO_PIN10_REG	LP GPIO10 configuration register	0x005C	varies
LP_GPIO_PIN11_REG	LP GPIO11 configuration register	0x0060	varies
LP_GPIO_PIN12_REG	LP GPIO12 configuration register	0x0064	varies
LP_GPIO_PIN13_REG	LP GPIO13 configuration register	0x0068	varies
LP_GPIO_PIN14_REG	LP GPIO14 configuration register	0x006C	varies
LP_GPIO_PIN15_REG	LP GPIO15 configuration register	0x0070	varies
LP_GPIO_FUNC0_IN_SEL_CFG_REG	Configuration register for input signal 0	0x0074	R/W
LP_GPIO_FUNC1_IN_SEL_CFG_REG	Configuration register for input signal 1	0x0078	R/W
LP_GPIO_FUNC2_IN_SEL_CFG_REG	Configuration register for input signal 2	0x007C	R/W
LP_GPIO_FUNC3_IN_SEL_CFG_REG	Configuration register for input signal 3	0x0080	R/W
LP_GPIO_FUNC4_IN_SEL_CFG_REG	Configuration register for input signal 4	0x0084	R/W
LP_GPIO_FUNC5_IN_SEL_CFG_REG	Configuration register for input signal 5	0x0088	R/W
LP_GPIO_FUNC6_IN_SEL_CFG_REG	Configuration register for input signal 6	0x008C	R/W
LP_GPIO_FUNC7_IN_SEL_CFG_REG	Configuration register for input signal 7	0x0090	R/W
LP_GPIO_FUNC8_IN_SEL_CFG_REG	Configuration register for input signal 8	0x0094	R/W
LP_GPIO_FUNC9_IN_SEL_CFG_REG	Configuration register for input signal 9	0x0098	R/W
LP_GPIO_FUNC10_IN_SEL_CFG_REG	Configuration register for input signal 10	0x009C	R/W
LP_GPIO_FUNC11_IN_SEL_CFG_REG	Configuration register for input signal 11	0x00A0	R/W
LP_GPIO_FUNC12_IN_SEL_CFG_REG	Configuration register for input signal 12	0x00A4	R/W
LP_GPIO_FUNC13_IN_SEL_CFG_REG	Configuration register for input signal 13	0x00A8	R/W
LP_GPIO_FUNC0_OUT_SEL_CFG_REG	Configuration register for LP GPIO0 output	0x00F4	R/W
LP_GPIO_FUNC1_OUT_SEL_CFG_REG	Configuration register for LP GPIO1 output	0x00F8	R/W
LP_GPIO_FUNC2_OUT_SEL_CFG_REG	Configuration register for LP GPIO2 output	0x00FC	R/W
LP_GPIO_FUNC3_OUT_SEL_CFG_REG	Configuration register for LP GPIO3 output	0x0100	R/W
LP_GPIO_FUNC4_OUT_SEL_CFG_REG	Configuration register for LP GPIO4 output	0x0104	R/W
LP_GPIO_FUNC5_OUT_SEL_CFG_REG	Configuration register for LP GPIO5 output	0x0108	R/W
LP_GPIO_FUNC6_OUT_SEL_CFG_REG	Configuration register for LP GPIO6 output	0x010C	R/W
LP_GPIO_FUNC7_OUT_SEL_CFG_REG	Configuration register for LP GPIO7 output	0x0110	R/W
LP_GPIO_FUNC8_OUT_SEL_CFG_REG	Configuration register for LP GPIO8 output	0x0114	R/W
LP_GPIO_FUNC9_OUT_SEL_CFG_REG	Configuration register for LP GPIO9 output	0x0118	R/W
LP_GPIO_FUNC10_OUT_SEL_CFG_REG	Configuration register for LP GPIO10 output	0x011C	R/W

Name	Description	Address	Access
LP_GPIO_FUNC11_OUT_SEL_CFG_REG	Configuration register for LP GPIO11 output	0x0120	R/W
LP_GPIO_FUNC12_OUT_SEL_CFG_REG	Configuration register for LP GPIO12 output	0x0124	R/W
LP_GPIO_FUNC13_OUT_SEL_CFG_REG	Configuration register for LP GPIO13 output	0x0128	R/W
LP_GPIO_FUNC14_OUT_SEL_CFG_REG	Configuration register for LP GPIO14 output	0x012C	R/W
LP_GPIO_FUNC15_OUT_SEL_CFG_REG	Configuration register for LP GPIO15 output	0x0130	R/W
Version Register			
LP_GPIO_VER_DATE_REG	Version control register	0x0004	R/W

9.19.5 LP IO MUX Register Summary

The addresses in this section are relative to LP IO MUX base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

The abbreviations given in Column **Access** are explained in Section *Access Types for Registers*.

Name	Description	Address	Access
Clock Enable Register			
LP_IOMUX_CLK_EN_REG	LP IO MUX clock gate register	0x0000	R/W
Configuration Registers			
LP_IOMUX_PAD0_REG	LP IO MUX configuration register for GPIO0	0x0008	R/W
LP_IOMUX_PAD1_REG	LP IO MUX configuration register for GPIO1	0x000C	R/W
LP_IOMUX_PAD2_REG	LP IO MUX configuration register for GPIO2	0x0010	R/W
LP_IOMUX_PAD3_REG	LP IO MUX configuration register for GPIO3	0x0014	R/W
LP_IOMUX_PAD4_REG	LP IO MUX configuration register for GPIO4	0x0018	R/W
LP_IOMUX_PAD5_REG	LP IO MUX configuration register for GPIO5	0x001C	R/W
LP_IOMUX_PAD6_REG	LP IO MUX configuration register for GPIO6	0x0020	R/W
LP_IOMUX_PAD7_REG	LP IO MUX configuration register for GPIO7	0x0024	R/W
LP_IOMUX_PAD8_REG	LP IO MUX configuration register for GPIO8	0x0028	R/W
LP_IOMUX_PAD9_REG	LP IO MUX configuration register for GPIO9	0x002C	R/W
LP_IOMUX_PAD10_REG	LP IO MUX configuration register for GPIO10	0x0030	R/W
LP_IOMUX_PAD11_REG	LP IO MUX configuration register for GPIO11	0x0034	R/W
LP_IOMUX_PAD12_REG	LP IO MUX configuration register for GPIO12	0x0038	R/W
LP_IOMUX_PAD13_REG	LP IO MUX configuration register for GPIO13	0x003C	R/W
LP_IOMUX_PAD14_REG	LP IO MUX configuration register for GPIO14	0x0040	R/W
LP_IOMUX_PAD15_REG	LP IO MUX configuration register for GPIO15	0x0044	R/W
LP_IOMUX_LP_PAD_HOLD_REG	LP GPIO PAD hold control register	0x004C	R/W
LP_IOMUX_LP_PAD_HYS_REG	LP GPIO PAD hysteresis control register	0x0050	R/W
Version Register			
LP_IOMUX_VER_DATE_REG	Version control register	0x0004	R/W

9.20 Registers

9.20.1 HP GPIO Matrix Registers

The addresses in this section are relative to HP GPIO matrix base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

For how to program reserved fields, please refer to Section IX .

Register 9.1. GPIO_OUT_REG (0x0004)

GPIO_OUT_DATA_ORIG	
31 … 0	
0x000000	Reset

GPIO_OUT_DATA_ORIG Configures the output value of GPIO0 ~ GPIO31 output in simple GPIO output mode.
0: Low level
1: High level
The value of bit0 ~ bit31 correspond to the output value of GPIO0 ~ GPIO31 respectively.
(R/W/SC/WTC)

Register 9.2. GPIO_OUT_W1TS_REG (0x0008)

GPIO_OUT_W1TS	
31 … 0	
0x000000	Reset

GPIO_OUT_W1TS Configures whether or not to set the output register GPIO_OUT_REG of GPIO0 ~ GPIO31.
0: Not set
1: The corresponding bit in GPIO_OUT_REG will be set to 1
Bit0 ~ bit31 are corresponding to GPIO0 ~ GPIO31. Recommended operation: use this register to set GPIO_OUT_REG.
(WT)

Register 9.3. GPIO_OUT_W1TC_REG (0x000C)

GPIO_OUT_W1TC	
31 … 0	
0x000000	Reset

GPIO_OUT_W1TC Configures whether or not to clear the output register GPIO_OUT_REG of GPIO0 ~ GPIO31 output.

0: Not clear

1: The corresponding bit in GPIO_OUT_REG will be cleared.

Bit0 ~ bit31 are corresponding to GPIO0 ~ GPIO31. Recommended operation: use this register to clear GPIO_OUT_REG.

(WT)

Register 9.4. GPIO_OUT1_REG (0x0010)

(reserved)	GPIO_OUT1_DATA_ORIG	
31 … 25	24 … 0	
0 0 0 0 0 0 0	0x00000	Reset

GPIO_OUT1_DATA_ORIG Configures the output value of GPIO32 ~ GPIO54 output in simple GPIO output mode.

0: Low level

1: High level

The value of bit0 ~ bit22 correspond to the output value of GPIO32 ~ GPIO54 respectively.

(R/W/SC/WTC)

Register 9.5. GPIO_OUT1_W1TS_REG (0x0014)

(reserved)							GPIO_OUT1_W1TS	
31						25	24	0
0	0	0	0	0	0	0	0x00000	Reset

GPIO_OUT1_W1TS Configures whether or not to set the output register GPIO_OUT1_REG of GPIO32 ~ GPIO54.

0: Not set

1: The corresponding bit in GPIO_OUT1_REG will be set to 1

Bit0 ~ bit22 are corresponding to GPIO32 ~ GPIO54. Recommended operation: use this register to set GPIO_OUT1_REG.

(WT)

Register 9.6. GPIO_OUT1_W1TC_REG (0x0018)

(reserved)							GPIO_OUT1_W1TC	
31						25	24	0
0	0	0	0	0	0	0	0x00000	Reset

GPIO_OUT1_W1TC Configures whether or not to clear the output register GPIO_OUT1_REG of GPIO32 ~ GPIO54 output.

0: Not clear

1: The corresponding bit in GPIO_OUT1_REG will be cleared.

Bit0 ~ bit22 are corresponding to GPIO32 ~ GPIO54. Recommended operation: use this register to clear GPIO_OUT_REG.

(WT)

Register 9.7. GPIO_ENABLE_REG (0x0020)

GPIO_ENABLE_DATA	
31 … 0	
0x000000	Reset

GPIO_ENABLE_DATA Configures whether or not to enable the output of GPIO0 ~ GPIO31.

0: Not enable

1: Enable

Bit0 ~ bit31 are corresponding to GPIO0 ~ GPIO31

(R/W/WTC)

Register 9.8. GPIO_ENABLE_W1TS_REG (0x0024)

GPIO_ENABLE_W1TS	
31 … 0	
0x000000	Reset

GPIO_ENABLE_W1TS Configures whether or not to set the output enable register GPIO_ENABLE_REG of GPIO0 ~ GPIO31.

0: Not set

1: The corresponding bit in GPIO_ENABLE_REG will be set to 1

Bit0 ~ bit31 are corresponding to GPIO0 ~ GPIO31. Recommended operation: use this register to set GPIO_ENABLE_REG.

(WT)

Register 9.9. GPIO_ENABLE_W1TC_REG (0x0028)

GPIO_ENABLE_W1TC	
31	0
0x000000	Reset

GPIO_ENABLE_W1TC Configures whether or not to clear the output enable register GPIO_ENABLE_REG of GPIO0 ~ GPIO31.

0: Not clear

1: The corresponding bit in GPIO_ENABLE_REG will be cleared

Bit0 ~ bit31 are corresponding to GPIO0 ~ 31. Recommended operation: use this register to clear GPIO_ENABLE_REG.

(WT)

Register 9.10. GPIO_ENABLE1_REG (0x002C)

(reserved)	GPIO_ENABLE1_DATA	
31 25	24 0	
0 0 0 0 0 0 0	0x00000	Reset

GPIO_ENABLE1_DATA Configures whether or not to enable the output of GPIO32 ~ GPIO54.

0: Not enable

1: Enable

Bit0 ~ bit22 are corresponding to GPIO32 ~ GPIO54.

(R/W/WTC)

Register 9.11. GPIO_ENABLE1_W1TS_REG (0x0030)

(reserved)	GPIO_ENABLE1_W1TS	
31 25	24 0	
0 0 0 0 0 0 0	0x00000	Reset

GPIO_ENABLE1_W1TS Configures whether or not to set the output enable register GPIO_ENABLE1_REG of GPIO32 ~ GPIO54.

0: Not set

1: The corresponding bit in GPIO_ENABLE1_REG will be set to 1

Bit0 ~ bit22 are corresponding to GPIO32 ~ GPIO54. Recommended operation: use this register to set GPIO_ENABLE1_REG.

(WT)

Register 9.12. GPIO_ENABLE1_W1TC_REG (0x0034)

(reserved)	GPIO_ENABLE1_W1TC	
31 25	24 0	
0 0 0 0 0 0 0	0x00000	Reset

GPIO_ENABLE1_W1TC Configures whether or not to clear the output enable register GPIO_ENABLE1_REG of GPIO32 ~ GPIO54.

0: Not clear

1: The corresponding bit in GPIO_ENABLE1_REG will be cleared

Bit0 ~ bit22 are corresponding to GPIO32 ~ 54. Recommended operation: use this register to clear GPIO_ENABLE1_REG.

(WT)

Register 9.13. GPIO_STRAP_REG (0x0038)

(reserved)	GPIO_STRAPPING	
31 ... 16	15 ... 0	
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	0x00	Reset

GPIO_STRAPPING Represents the values of GPIO strapping pins.

bit0: GPIO38

bit1: GPIO37

bit2: GPIO36

bit3: GPIO35

bit4: GPIO34

bit5 ~ bit15: invalid

For more information about the functions controlled by strapping pins, see *ESP32-P4 Datasheet* > Section *Strapping Pins*. (RO)

Register 9.14. GPIO_IN_REG (0x003C)

GPIO_IN_DATA_NEXT	
31 ... 0	
0x000000	Reset

GPIO_IN_DATA_NEXT Represents the input value of GPIO0 ~ GPIO31. Each bit represents a pin input value:

0: Low level

1: High level

Bit0 ~ bit31 are corresponding to GPIO0 ~ GPIO31.

(RO)

Register 9.15. GPIO_IN1_REG (0x0040)

(reserved)							GPIO_IN1_DATA_NEXT
31						25	24 ... 0
0	0	0	0	0	0	0	0x00000

Reset

GPIO_IN1_DATA_NEXT Represents the input value of GPIO32 ~ GPIO54. Each bit represents a pin input value:
0: Low level
1: High level
Bit0 ~ bit22 are corresponding to GPIO32 ~ GPIO54.
(RO)

Register 9.16. GPIO_STATUS_REG (0x0044)

GPIO_STATUS_INTERRUPT
31 ... 0
0x000000

Reset

GPIO_STATUS_INTERRUPT The interrupt status of GPIO0 ~ GPIO31, can be configured by the software.
Bit0 ~ bit31 are corresponding to GPIO0 ~ GPIO31.
Each bit represents the status of its corresponding GPIO:
0: Represents the GPIO does not generate the interrupt configured by GPIO_PIN*n*_INT_TYPE, or this bit is configured to 0 by the software.
1: Represents the GPIO generates the interrupt configured by GPIO_PIN*n*_INT_TYPE, or this bit is configured to 1 by the software.
(R/W/WTC)

Register 9.17. GPIO_STATUS_W1TS_REG (0x0048)

GPIO_STATUS_W1TS	
31 … 0	
0x000000	Reset

GPIO_STATUS_W1TS Configures whether or not to set the interrupt status register GPIO_STATUS_INTERRUPT of GPIO0 ~ GPIO31.
Bit0 ~ bit31 are corresponding to GPIO0 ~ GPIO31.
If the value 1 is written to a bit here, the corresponding bit in GPIO_STATUS_INTERRUPT will be set to 1.
Recommended operation: use this register to set GPIO_STATUS_INTERRUPT.
(WT)

Register 9.18. GPIO_STATUS_W1TC_REG (0x004C)

GPIO_STATUS_W1TC	
31 … 0	
0x000000	Reset

GPIO_STATUS_W1TC Configures whether or not to clear the interrupt status register GPIO_STATUS_INTERRUPT of GPIO0 ~ GPIO31.
Bit0 ~ bit31 are corresponding to GPIO0 ~ GPIO31.
If the value 1 is written to a bit here, the corresponding bit in GPIO_STATUS_INTERRUPT will be cleared.
Recommended operation: use this register to clear GPIO_STATUS_INTERRUPT.
(WT)

Register 9.19. GPIO_STATUS1_REG (0x0050)

(reserved)	GPIO_STATUS1_INTERRUPT	
31 25	24 0	
0 0 0 0 0 0 0	0x00000	Reset

GPIO_STATUS1_INTERRUPT The interrupt status of GPIO32 ~ GPIO54, can be configured by the software.
Bit0 ~ bit22 are corresponding to GPIO32 ~ GPIO54.
Each bit represents the status of its corresponding GPIO:
0: Represents the GPIO does not generate the interrupt configured by GPIO_PIN*n*_INT_TYPE, or this bit is configured to 0 by the software.
1: Represents the GPIO generates the interrupt configured by GPIO_PIN*n*_INT_TYPE, or this bit is configured to 1 by the software.
(R/W/WTC)

Register 9.20. GPIO_STATUS1_W1TS_REG (0x0054)

(reserved)	GPIO_STATUS1_W1TS	
31 25	24 0	
0 0 0 0 0 0 0	0x00000	Reset

GPIO_STATUS1_W1TS Configures whether or not to set the interrupt status register GPIO_STATUS1_INTERRUPT of GPIO32 ~ GPIO54.
Bit0 ~ bit22 are corresponding to GPIO32 ~ GPIO54.
If the value 1 is written to a bit here, the corresponding bit in GPIO_STATUS1_INTERRUPT will be set to 1.
Recommended operation: use this register to set GPIO_STATUS1_INTERRUPT.
(WT)

Register 9.21. GPIO_STATUS1_W1TC_REG (0x0058)

(reserved)	GPIO_STATUS1_W1TC	
31 … 25	24 … 0	
0 0 0 0 0 0 0	0x00000	Reset

GPIO_STATUS1_W1TC Configures whether or not to clear the interrupt status register GPIO_STATUS1_INTERRUPT of GPIO32 ~ GPIO54.

Bit0 ~ bit22 are corresponding to GPIO32 ~ GPIO54.

If the value 1 is written to a bit here, the corresponding bit in GPIO_STATUS1_INTERRUPT will be cleared.

Recommended operation: use this register to clear GPIO_STATUS1_INTERRUPT.

(WT)

Register 9.22. GPIO_INTR_0_REG (0x005C)

GPIO_INT_0	
31 … 0	
0x000000	Reset

GPIO_INT_0 Represents the CPU interrupt 0 status of GPIO0 ~ GPIO31. Each bit represents:

0: Represents CPU interrupt is not enabled, or the GPIO does not generate the interrupt configured by GPIO_PIN*n*_INT_TYPE.

1: Represents the GPIO generates an interrupt configured by GPIO_PIN*n*_INT_TYPE after the CPU interrupt is enabled.

Bit0 ~ bit31 are corresponding to GPIO0 ~ GPIO31. This interrupt status is corresponding to the bit in GPIO_STATUS_REG when assert (high) enable signal (bit13 of GPIO_PIN*n*_REG).

(RO)

Register 9.23. GPIO_INTR1_0_REG (0x0060)

(reserved)	GPIO_INT1_0	
31 … 25	24 … 0	
0 0 0 0 0 0 0	0x00000	Reset

GPIO_INT1_0 Represents the CPU interrupt 0 status of GPIO32 ~ GPIO54. Each bit represents:

0: Represents CPU interrupt is not enabled, or the GPIO does not generate the interrupt configured by GPIO_PIN*n*_INT_TYPE.

1: Represents the GPIO generates an interrupt configured by GPIO_PIN*n*_INT_TYPE after the CPU interrupt is enabled.

Bit0 ~ bit22 are corresponding to GPIO32 ~ GPIO54. This interrupt status is corresponding to the bit in GPIO_STATUS1_REG when assert (high) enable signal (bit13 of GPIO_PIN*n*_REG).

(RO)

Register 9.24. GPIO_INTR_1_REG (0x0064)

GPIO_INT_1	
31 … 0	
0x000000	Reset

GPIO_INT_1 Represents the CPU interrupt 1 status of GPIO0 ~ GPIO31. Each bit represents:

0: Represents CPU interrupt is not enabled, or the GPIO does not generate the interrupt configured by GPIO_PIN*n*_INT_TYPE.

1: Represents the GPIO generates an interrupt configured by GPIO_PIN*n*_INT_TYPE after the CPU interrupt is enabled.

Bit0 ~ bit31 are corresponding to GPIO0 ~ GPIO31. This interrupt status is corresponding to the bit in GPIO_STATUS_REG when assert (high) enable signal (bit14 of GPIO_PIN*n*_REG).

(RO)

Register 9.25. GPIO_INTR1_1_REG (0x0068)

(reserved)	GPIO_INT1_1	
31 … 25	24 … 0	
0 0 0 0 0 0 0	0x00000	Reset

GPIO_INT1_1 Represents the CPU interrupt 1 status of GPIO32 ~ GPIO54. Each bit represents:

0: Represents CPU interrupt is not enabled, or the GPIO does not generate the interrupt configured by GPIO_PIN*n*_INT_TYPE.

1: Represents the GPIO generates an interrupt configured by GPIO_PIN*n*_INT_TYPE after the CPU interrupt is enabled.

Bit0 ~ bit22 are corresponding to GPIO32 ~ GPIO54. This interrupt status is corresponding to the bit in GPIO_STATUS1_REG when assert (high) enable signal (bit14 of GPIO_PIN*n*_REG).

(RO)

Register 9.26. GPIO_STATUS_NEXT_REG (0x006C)

GPIO_STATUS_INTERRUPT_NEXT	
31 … 0	
0x000000	Reset

GPIO_STATUS_INTERRUPT_NEXT Represents the interrupt source signal of GPIO0 ~ GPIO31.

Bit0 ~ bit31 are corresponding to GPIO0 ~ GPIO31. Each bit represents:

0: The GPIO does not generate the interrupt configured by GPIO_PIN*n*_INT_TYPE.

1: The GPIO generates an interrupt configured by GPIO_PIN*n*_INT_TYPE.

The interrupt could be rising edge interrupt, falling edge interrupt, level sensitive interrupt and any edge interrupt.

(RO)

Register 9.27. GPIO_STATUS_NEXT1_REG (0x0070)

(reserved)							GPIO_STATUS_INTERRUPT_NEXT1	
31						25	24 0	
0	0	0	0	0	0	0	0x00000	Reset

GPIO_STATUS_INTERRUPT_NEXT1 Represents the interrupt source signal of GPIO32 ~ GPIO54.
Bit0 ~ bit22 are corresponding to GPIO32 ~ GPIO54. Each bit represents:
0: The GPIO does not generate the interrupt configured by GPIO_PIN*n*_INT_TYPE.
1: The GPIO generates an interrupt configured by GPIO_PIN*n*_INT_TYPE.
The interrupt could be rising edge interrupt, falling edge interrupt, level sensitive interrupt and any edge interrupt.
(RO)

Register 9.28. GPIO_PINn_REG (n: 0 - 54) (0x0074+0x4*n)

(reserved)	GPIO_PINn_INT_ENA	(reserved)	GPIO_PINn_WAKEUP_ENABLE	GPIO_PINn_INT_TYPE	(reserved)	GPIO_PINn_SYNC1_BYPASS	GPIO_PINn_PAD_DRIVER	GPIO_PINn_SYNC2_BYPASS	
31 … 18	17 … 13	12 … 11	10	9 … 7	6 … 5	4 … 3	2	1 … 0	
0 0 0 0 0 0 0 0 0 0 0 0 0 0	0x0	0x0	0	0x0	0 0	0x0	0	0x0	Reset

GPIO_PINn_SYNC2_BYPASS Configures whether or not to synchronize GPIO input data on either edge of HP IO MUX operating clock for the second-level synchronization.
0: Not synchronize
1: Synchronize on falling edge
2: Synchronize on rising edge
3: Synchronize on rising edge
(R/W)

GPIO_PINn_PAD_DRIVER Configures to select pin drive mode.
0: Normal output
1: Open drain output
(R/W)

GPIO_PINn_SYNC1_BYPASS Configures whether or not to synchronize GPIO input data on either edge of HP IO MUX operating clock for the first-level synchronization.
0: Not synchronize
1: Synchronize on falling edge
2: Synchronize on rising edge
3: Synchronize on rising edge
(R/W)

GPIO_PINn_INT_TYPE Configures GPIO interrupt type.
0: GPIO interrupt disabled
1: Rising edge trigger
2: Falling edge trigger
3: Any edge trigger
4: Low level trigger
5: High level trigger
(R/W)

GPIO_PINn_WAKEUP_ENABLE Configures whether or not to enable GPIO wake-up function.
0: Disable
1: Enable
This function only wakes up the CPU from Light-sleep.
(R/W)

Continued on the next page...

Register 9.28. GPIO_PIN*n*_REG (*n*: 0 - 54) (0x0074+0x4*n)

Continued from the previous page...

GPIO_PIN*n*_INT_ENA Configures whether or not to enable CPU interrupt or CPU non-maskable interrupt.

bit13: Configures whether or not to enable CPU interrupt 0:
0: Disable
1: Enable
bit14: Configures whether or not to enable CPU interrupt 1:
0: Disable
1: Enable
bit15: invalid
bit16: Configures whether or not to enable CPU interrupt 2:
0: Disable
1: Enable
bit17: Configures whether or not to enable CPU interrupt 3:
0: Disable
1: Enable
(R/W)

Register 9.29. GPIO_FUNCn_IN_SEL_CFG_REG (n: 0 - 255) (0x015C+4*n)

(reserved)	GPIO_SIGn_IN_SEL	GPIO_FUNCn_IN_INV_SEL	GPIO_FUNCn_IN_SEL	
31 8	7	6	5 0	
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	0	0	0x3f	Reset

GPIO_FUNCn_IN_SEL Configures to select a pin from the 55 GPIO pins to connect the input signal *n*.

0: Select GPIO0

1: Select GPIO1

......

53: Select GPIO53

54: Select GPIO54

55 ~ 61: invalid

Or

62: A constantly low input

63: A constantly high input

(R/W)

GPIO_FUNCn_IN_INV_SEL Configures whether or not to invert the input value.

0: Not invert

1: Invert

(R/W)

GPIO_SIGn_IN_SEL Configures whether or not to route signals via HP GPIO matrix.

0: Bypass HP GPIO matrix, i.e., connect signals directly to peripheral configured in HP IO MUX.

1: Route signals via HP GPIO matrix.

(R/W)

Register 9.30. GPIO_FUNCn_OUT_SEL_CFG_REG (n: 0 - 54) (0x0558+0x4*n)

	(reserved)	GPIO_FUNCn_OE_INV_SEL	GPIO_FUNCn_OE_SEL	GPIO_FUNCn_OUT_INV_SEL	GPIO_FUNCn_OUT_SEL	
Bits	31 … 12	11	10	9	8 … 0	
	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	0	0	0	0x100	Reset

GPIO_FUNCn_OUT_SEL Configures to select a signal *Y* (0 <= *Y* < 256) from 256 peripheral signals to be output to GPIO*n*.

0: Select signal 0

1: Select signal 1

......

254: Select signal 254

255: Select signal 255

Or

256: Bit*n* of GPIO_OUT_REG and GPIO_OUT1_REG are selected as the output value, GPIO_ENABLE_REG and GPIO_ENABLE1_REG are selected as the output enable.

257 ~ 511: invalid

For the detailed signal list, see Table 9.12-1.

(R/W/SC)

GPIO_FUNCn_OUT_INV_SEL Configures whether or not to invert the output value.

0: Not invert

1: Invert

(R/W/SC)

GPIO_FUNCn_OE_SEL Configures to select the source of output enable signal.

0: Use output enable signal from peripheral.

1: Force the output enable signal to be sourced from bit*n* (*n*: 0 ~ 31) of GPIO_ENABLE_REG and bit*n*-32 (*n*: 32 ~ 54) of GPIO_ENABLE1_REG.

(R/W)

GPIO_FUNCn_OE_INV_SEL Configures whether or not to invert the output enable signal.

0: Not invert

1: Invert

(R/W)

Register 9.31. GPIO_INTR_2_REG (0x063C)

GPIO_INT_2	
31 … 0	
0x000000	Reset

GPIO_INT_2 Represents the CPU interrupt 2 status of GPIO0 ~ GPIO31. Each bit represents:
0: Represents CPU interrupt is not enabled, or the GPIO does not generate the interrupt configured by GPIO_PIN*n*_INT_TYPE.
1: Represents the GPIO generates an interrupt configured by GPIO_PIN*n*_INT_TYPE after the CPU interrupt is enabled.
Bit0 ~ bit31 are corresponding to GPIO0 ~ GPIO31. This interrupt status is corresponding to the bit in GPIO_STATUS_REG when assert (high) enable signal (bit16 of GPIO_PIN*n*_REG).
(RO)

Register 9.32. GPIO_INTR1_2_REG (0x0640)

(reserved)	GPIO_INT1_2	
31 … 25	24 … 0	
0 0 0 0 0 0 0	0x00000	Reset

GPIO_INT1_2 Represents the CPU interrupt 2 status of GPIO32 ~ GPIO54. Each bit represents:
0: Represents CPU interrupt is not enabled, or the GPIO does not generate the interrupt configured by GPIO_PIN*n*_INT_TYPE.
1: Represents the GPIO generates an interrupt configured by GPIO_PIN*n*_INT_TYPE after the CPU interrupt is enabled.
Bit0 ~ bit22 are corresponding to GPIO32 ~ GPIO54. This interrupt status is corresponding to the bit in GPIO_STATUS1_REG when assert (high) enable signal (bit16 of GPIO_PIN*n*_REG).
(RO)

Register 9.33. GPIO_INTR_3_REG (0x0644)

GPIO_INT_3	
31 ... 0	
0x000000	Reset

GPIO_INT_3 Represents the CPU interrupt 3 status of GPIO0 ~ GPIO31. Each bit represents:

0: Represents CPU interrupt is not enabled, or the GPIO does not generate the interrupt configured by GPIO_PIN*n*_INT_TYPE.

1: Represents the GPIO generates an interrupt configured by GPIO_PIN*n*_INT_TYPE after the CPU interrupt is enabled.

Bit0 ~ bit31 are corresponding to GPIO0 ~ GPIO31. This interrupt status is corresponding to the bit in GPIO_STATUS_REG when assert (high) enable signal (bit17 of GPIO_PIN*n*_REG).

(RO)

Register 9.34. GPIO_INTR1_3_REG (0x0648)

(reserved)	GPIO_INT1_3	
31 ... 25	24 ... 0	
0 0 0 0 0 0 0	0x00000	Reset

GPIO_INT1_3 Represents the CPU interrupt 3 status of GPIO32 ~ GPIO54. Each bit represents:

0: Represents CPU interrupt is not enabled, or the GPIO does not generate the interrupt configured by GPIO_PIN*n*_INT_TYPE.

1: Represents the GPIO generates an interrupt configured by GPIO_PIN*n*_INT_TYPE after the CPU interrupt is enabled.

Bit0 ~ bit22 are corresponding to GPIO32 ~ GPIO54. This interrupt status is corresponding to the bit in GPIO_STATUS1_REG when assert (high) enable signal (bit17 of GPIO_PIN*n*_REG).

Register 9.35. GPIO_CLOCK_GATE_REG (0x064C)

(reserved)	GPIO_CLK_EN	
31 ... 1	0	
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	1	Reset

GPIO_CLK_EN Configures whether or not to enable clock gate.

0: Not enable

1: Enable, the clock is free running.

(R/W)

Register 9.36. GPIO_ZERO_DET0_FILTER_CNT_REG (0x0710)

GPIO_ZERO_DET0_FILTER_CNT	
31 … 0	
0xffffffff	Reset

GPIO_ZERO_DET0_FILTER_CNT Configures to change GPIO analog comparator0 zero detect filter's threshold.
Interrupts in GPIO_ZERO_DET0_FILTER_CNT iomux_core_clk cycles after one interrupt will be fillterred, and not be seen by CPU. (R/W)

Register 9.37. GPIO_ZERO_DET1_FILTER_CNT_REG (0x0714)

GPIO_ZERO_DET1_FILTER_CNT	
31 … 0	
0xffffffff	Reset

GPIO_ZERO_DET1_FILTER_CNT Configures to change GPIO analog comparator1 zero detect filter's threshold.
Interrupts in GPIO_ZERO_DET0_FILTER_CNT iomux_core_clk cycles after one interrupt will be fillterred, and not be seen by CPU. (R/W)

Register 9.38. GPIO_DATE_REG (0x07FC)

(reserved)	GPIO_DATE	
31 … 28	27 … 0	
0 0 0 0	0x230403	Reset

GPIO_DATE Version control register.
(R/W)

Register 9.39. GPIO_INT_RAW_REG (0x0700)

	(reserved)	GPIO_COMP1_ALL_INT_RAW	GPIO_COMP1_POS_INT_RAW	GPIO_COMP1_NEG_INT_RAW	GPIO_COMP0_ALL_INT_RAW	GPIO_COMP0_POS_INT_RAW	GPIO_COMP0_NEG_INT_RAW
Bit	31 ... 6	5	4	3	2	1	0
Reset	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	0	0	0	0	0	0

GPIO_COMP0_NEG_INT_RAW The raw interrupt status of GPIO_COMP0_NEG_INT. (R/WTC/SS)

GPIO_COMP0_POS_INT_RAW The raw interrupt status of GPIO_COMP0_POS_INT. (R/WTC/SS)

GPIO_COMP0_ALL_INT_RAW The raw interrupt status of GPIO_COMP0_ALL_INT. (R/WTC/SS)

GPIO_COMP1_NEG_INT_RAW The raw interrupt status of GPIO_COMP1_NEG_INT. (R/WTC/SS)

GPIO_COMP1_POS_INT_RAW The raw interrupt status of GPIO_COMP1_POS_INT. (R/WTC/SS)

GPIO_COMP1_ALL_INT_RAW The raw interrupt status of GPIO_COMP1_ALL_INT. (R/WTC/SS)

Register 9.40. GPIO_INT_ST_REG (0x0704)

	(reserved)	GPIO_COMP1_ALL_INT_ST	GPIO_COMP1_POS_INT_ST	GPIO_COMP1_NEG_INT_ST	GPIO_COMP0_ALL_INT_ST	GPIO_COMP0_POS_INT_ST	GPIO_COMP0_NEG_INT_ST
Bit	31 ... 6	5	4	3	2	1	0
Reset	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	0	0	0	0	0	0

GPIO_COMP0_NEG_INT_ST The masked interrupt status of GPIO_COMP0_NEG_INT. (RO)

GPIO_COMP0_POS_INT_ST The masked interrupt status of GPIO_COMP0_POS_INT. (RO)

GPIO_COMP0_ALL_INT_ST The masked interrupt status of GPIO_COMP0_ALL_INT. (RO)

GPIO_COMP1_NEG_INT_ST The masked interrupt status of GPIO_COMP1_NEG_INT. (RO)

GPIO_COMP1_POS_INT_ST The masked interrupt status of GPIO_COMP1_POS_INT. (RO)

GPIO_COMP1_ALL_INT_ST The masked interrupt status of GPIO_COMP1_ALL_INT. (RO)

Register 9.41. GPIO_INT_ENA_REG (0x0708)

	(reserved)	GPIO_COMP1_ALL_INT_ENA	GPIO_COMP1_POS_INT_ENA	GPIO_COMP1_NEG_INT_ENA	GPIO_COMP0_ALL_INT_ENA	GPIO_COMP0_POS_INT_ENA	GPIO_COMP0_NEG_INT_ENA
Bits	31 … 6	5	4	3	2	1	0
Reset	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	1	1	1	1	1	1

GPIO_COMP0_NEG_INT_ENA Write 1 to enable GPIO_COMP0_NEG_INT. (R/W)

GPIO_COMP0_POS_INT_ENA Write 1 to enable GPIO_COMP0_POS_INT. (R/W)

GPIO_COMP0_ALL_INT_ENA Write 1 to enable GPIO_COMP0_ALL_INT. (R/W)

GPIO_COMP1_NEG_INT_ENA Write 1 to enable GPIO_COMP1_NEG_INT. (R/W)

GPIO_COMP1_POS_INT_ENA Write 1 to enable GPIO_COMP1_POS_INT. (R/W)

GPIO_COMP1_ALL_INT_ENA Write 1 to enable GPIO_COMP1_ALL_INT. (R/W)

Register 9.42. GPIO_INT_CLR_REG (0x070C)

	(reserved)	GPIO_COMP1_ALL_INT_CLR	GPIO_COMP1_POS_INT_CLR	GPIO_COMP1_NEG_INT_CLR	GPIO_COMP0_ALL_INT_CLR	GPIO_COMP0_POS_INT_CLR	GPIO_COMP0_NEG_INT_CLR
Bits	31 … 6	5	4	3	2	1	0
Reset	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	0	0	0	0	0	0

GPIO_COMP0_NEG_INT_CLR Write 1 to clear GPIO_COMP0_NEG_INT. (WT)

GPIO_COMP0_POS_INT_CLR Write 1 to clear GPIO_COMP0_POS_INT. (WT)

GPIO_COMP0_ALL_INT_CLR Write 1 to clear GPIO_COMP0_ALL_INT. (WT)

GPIO_COMP1_NEG_INT_CLR Write 1 to clear GPIO_COMP1_NEG_INT. (WT)

GPIO_COMP1_POS_INT_CLR Write 1 to clear GPIO_COMP1_POS_INT. (WT)

GPIO_COMP1_ALL_INT_CLR Write 1 to clear GPIO_COMP1_ALL_INT. (WT)

9.20.2 HP IO MUX Registers

The addresses in this section are relative to the HP IO MUX base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

For how to program reserved fields, please refer to Section IX .

Register 9.43. IO_MUX_GPIO*n*_REG (*n*: 0 - 54) (0x0004+4n*)**

(reserved)	IO_MUX_GPIO*n*_FILTER_EN	IO_MUX_GPIO*n*_MCU_SEL	IO_MUX_GPIO*n*_FUN_DRV	IO_MUX_GPIO*n*_FUN_IE	IO_MUX_GPIO*n*_FUN_WPU	IO_MUX_GPIO*n*_FUN_WPD	IO_MUX_GPIO*n*_MCU_DRV	IO_MUX_GPIO*n*_MCU_IE	IO_MUX_GPIO*n*_MCU_WPU	IO_MUX_GPIO*n*_MCU_WPD	IO_MUX_GPIO*n*_SLP_SEL	IO_MUX_GPIO*n*_MCU_OE	
31 … 16	15	14 … 12	11 … 10	9	8	7	6 … 5	4	3	2	1	0	
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	0	0x0	0x2	0	0	0	0x0	0	0	0	0	0	Reset

IO_MUX_GPIO*n*_MCU_OE Configures whether or not to enable the output of GPIO*n* in sleep mode.
0: Disable
1: Enable
(R/W)

IO_MUX_GPIO*n*_SLP_SEL Configures whether or not to enter sleep mode for GPIO*n*.
0: Not enter
1: Enter
(R/W)

IO_MUX_GPIO*n*_MCU_WPD Configure whether or not to enable pull-down resistor of GPIO*n* during sleep mode.
0: Disable
1: Enable
(R/W)

IO_MUX_GPIO*n*_MCU_WPU Configures whether or not to enable pull-up resistor of GPIO*n* during sleep mode.
0: Disable
1: Enable
(R/W)

IO_MUX_GPIO*n*_MCU_IE Configures whether or not to enable the input of GPIO*n* during sleep mode.
0: Disable
1: Enable
(R/W)

IO_MUX_GPIO*n*_MCU_DRV Configures the drive strength of GPIO*n* during sleep mode.
0: ~5 mA
1: ~10 mA
2: ~20 mA
3: ~40 mA
(R/W)

Continued on the next page...

Register 9.43. IO_MUX_GPIOn_REG (n: 0 - 54) (0x0004+4*n)

Continued from the previous page...

IO_MUX_GPIOn_FUN_WPD Configures whether or not to enable pull-down resistor of GPIOn.
0: Disable
1: Enable
(R/W)

IO_MUX_GPIOn_FUN_WPU Configures whether or not enable pull-up resistor of GPIOn.
0: Disable
1: Enable
(R/W)

IO_MUX_GPIOn_FUN_IE Configures whether or not to enable input of GPIOn.
0: Disable
1: Enable
(R/W)

IO_MUX_GPIOn_FUN_DRV Configures the drive strength of GPIOn.
0: ~5 mA
1: ~10 mA
2: ~20 mA
3: ~40 mA
(R/W)

IO_MUX_GPIOn_MCU_SEL Configures to select IO MUX function for this pin.
0: Select Function 0
1: Select Function 1
......
(R/W)

IO_MUX_GPIOn_FILTER_EN Configures whether or not to enable filter for pin input signals.
0: Disable
1: Enable
(R/W)

Register 9.44. IO_MUX_DATE_REG (0x0104)

(reserved)	IO_MUX_DATE	
31 28	27 0	
0 0 0 0	0x201222	Reset

IO_MUX_DATE Version control register.
(R/W)

9.20.3 GPIO EXT Registers

The addresses in this section are relative to (HP GPIO matrix base address + 0x0F00). GPIO base address is provided in Table 7.3-2 in Chapter 7 *System and Memory*.

For how to program reserved fields, please refer to Section IX .

Register 9.45. GPIO_EXT_SIGMADELTAn_REG (n: 0 - 7) (0x0000+0x4*n)

	(reserved)	GPIO_EXT_SDn_PRESCALE	GPIO_EXT_SDn_IN	
Bits	31 … 16	15 … 8	7 … 0	
	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	0xff	0x0	Reset

GPIO_EXT_SDn_IN (n: 0 - 7) Configures the duty cycle of sigma delta modulation output.
(R/W)

GPIO_EXT_SDn_PRESCALE (n: 0 - 7) Configures the divider value to divide HP IO MUX operating clock. Can be 0 ~ 255.
0: Do not divide
1 ~ 255: The clock is divided by 2 ~ 256
(R/W)

Register 9.46. GPIO_EXT_SIGMADELTA_MISC_REG (0x0024)

	(reserved)	GPIO_EXT_SD_FUNCTION_CLK_EN	(reserved)	
Bits	31	30	29 … 0	
	0	0	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	Reset

GPIO_EXT_SD_FUNCTION_CLK_EN Configures whether or not to enable the clock for sigma delta modulation.
0: Not enable
1: Enable
(R/W)

Register 9.47. GPIO_EXT_GLITCH_FILTER_CHn_REG (n: 0 - 7) (0x0030+0x4*n)

(reserved)	GPIO_EXT_FILTER_CHn_WINDOW_WIDTH	GPIO_EXT_FILTER_CHn_WINDOW_THRES	GPIO_EXT_FILTER_CHn_INPUT_IO_NUM	GPIO_EXT_FILTER_CHn_EN	
31 19	18 13	12 7	6 1	0	
0 0 0 0 0 0 0 0 0 0 0 0 0	0x0	0x0	0x0	0	Reset

GPIO_EXT_FILTER_CHn_EN Configures whether or not to enable channel *n* of Glitch Filter.
0: Not enable
1: Enable
(R/W)

GPIO_EXT_FILTER_CHn_INPUT_IO_NUM Configures to select the input GPIO for Glitch Filter.
0: Select GPIO0
1: Select GPIO1
......
53: Select GPIO53
54: Select GPIO54
(R/W)

GPIO_EXT_FILTER_CHn_WINDOW_THRES Configures the window threshold for Glitch Filter. The window threshold should be less than or equal to GPIO_EXT_FILTER_CHn_WINDOW_WIDTH.
Measurement unit: HP IO MUX operating clock cycle
(R/W)

GPIO_EXT_FILTER_CHn_WINDOW_WIDTH Configures the window width for Glitch Filter. The effective value of window width is 0 ~ 62. 63 is a reserved value and can not be used.
Measurement unit: HP IO MUX operating clock cycle
(R/W)

Register 9.48. GPIO_EXT_ETM_EVENT_CHn_CFG_REG (n: 0 - 7) (0x0060+0x4*n)

(reserved)	GPIO_EXT_ETM_CH0_EVENT_EN	(reserved)	GPIO_EXT_ETM_CH0_EVENT_SEL	
31 … 8	7	6	5 … 0	
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	0	0	0x0	Reset

GPIO_EXT_ETM_CHn_EVENT_SEL Configures to select GPIO for ETM event channel.

0: Select GPIO0

1: Select GPIO1

......

53: Select GPIO53

54: Select GPIO54

(R/W)

GPIO_EXT_ETM_CHn_EVENT_EN Configures whether or not to enable ETM event send.

0: Not enable

1: Enable

(R/W)

Register 9.49. GPIO_EXT_ETM_TASK_P0_CFG_REG (0x00A0)

(reserved)	GPIO_EXT_ETM_TASK_GPIO3_SEL	GPIO_EXT_ETM_TASK_GPIO3_EN	(reserved)	GPIO_EXT_ETM_TASK_GPIO2_SEL	GPIO_EXT_ETM_TASK_GPIO2_EN	(reserved)	GPIO_EXT_ETM_TASK_GPIO1_SEL	GPIO_EXT_ETM_TASK_GPIO1_EN	(reserved)	GPIO_EXT_ETM_TASK_GPIO0_SEL	GPIO_EXT_ETM_TASK_GPIO0_EN	
31 28	27 25	24	23 20	19 17	16	15 12	11 9	8	7 4	3 1	0	
0 0 0 0	0x0	0	0 0 0 0	0x0	0	0 0 0 0	0x0	0	0 0 0 0	0x0	0	Reset

GPIO_EXT_ETM_TASK_GPIO*n*_EN (*n*: 0 - 3) Configures whether or not to enable GPIO*n* to response ETM task.

0: Not enable

1: Enable

(R/W)

GPIO_EXT_ETM_TASK_GPIO*n*_SEL (*n*: 0 - 3) Configures to select an ETM task channel for GPIO*n*.

0: Select channel 0

1: Select channel 1

......

7: Select channel 7

(R/W)

Register 9.50. GPIO_EXT_ETM_TASK_P1_CFG_REG (0x00A4)

(reserved)	GPIO_EXT_ETM_TASK_GPIO7_SEL	GPIO_EXT_ETM_TASK_GPIO7_EN	(reserved)	GPIO_EXT_ETM_TASK_GPIO6_SEL	GPIO_EXT_ETM_TASK_GPIO6_EN	(reserved)	GPIO_EXT_ETM_TASK_GPIO5_SEL	GPIO_EXT_ETM_TASK_GPIO5_EN	(reserved)	GPIO_EXT_ETM_TASK_GPIO4_SEL	GPIO_EXT_ETM_TASK_GPIO4_EN	
31 28	27 25	24	23 20	19 17	16	15 12	11 9	8	7 4	3 1	0	
0 0 0 0	0x0	0	0 0 0 0	0x0	0	0 0 0 0	0x0	0	0 0 0 0	0x0	0	Reset

GPIO_EXT_ETM_TASK_GPIO*n*_EN (*n*: 4 - 7) Configures whether or not to enable GPIO*n* to response ETM task.

0: Not enable

1: Enable

(R/W)

GPIO_EXT_ETM_TASK_GPIO*n*_SEL (*n*: 4 - 7) Configures to select an ETM task channel for GPIO*n*.

0: Select channel 0

1: Select channel 1

......

7: Select channel 7

(R/W)

Register 9.51. GPIO_EXT_ETM_TASK_P2_CFG_REG (0x00A8)

(reserved)	GPIO_EXT_ETM_TASK_GPIO11_SEL	GPIO_EXT_ETM_TASK_GPIO11_EN	(reserved)	GPIO_EXT_ETM_TASK_GPIO10_SEL	GPIO_EXT_ETM_TASK_GPIO10_EN	(reserved)	GPIO_EXT_ETM_TASK_GPIO9_SEL	GPIO_EXT_ETM_TASK_GPIO9_EN	(reserved)	GPIO_EXT_ETM_TASK_GPIO8_SEL	GPIO_EXT_ETM_TASK_GPIO8_EN	
31 28	27 25	24	23 20	19 17	16	15 12	11 9	8	7 4	3 1	0	
0 0 0 0	0x0	0	0 0 0 0	0x0	0	0 0 0 0	0x0	0	0 0 0 0	0x0	0	Reset

GPIO_EXT_ETM_TASK_GPIO*n*_EN (*n*: 8 - 11) Configures whether or not to enable GPIO*n* to response ETM task.

0: Not enable

1: Enable

(R/W)

GPIO_EXT_ETM_TASK_GPIO*n*_SEL (*n*: 8 - 11) Configures to select an ETM task channel for GPIO*n*.

0: Select channel 0

1: Select channel 1

......

7: Select channel 7

(R/W)

Register 9.52. GPIO_EXT_ETM_TASK_P3_CFG_REG (0x00AC)

(reserved)	GPIO_EXT_ETM_TASK_GPIO15_SEL	GPIO_EXT_ETM_TASK_GPIO15_EN	(reserved)	GPIO_EXT_ETM_TASK_GPIO14_SEL	GPIO_EXT_ETM_TASK_GPIO14_EN	(reserved)	GPIO_EXT_ETM_TASK_GPIO13_SEL	GPIO_EXT_ETM_TASK_GPIO13_EN	(reserved)	GPIO_EXT_ETM_TASK_GPIO12_SEL	GPIO_EXT_ETM_TASK_GPIO12_EN	
31 28	27 25	24	23 20	19 17	16	15 12	11 9	8	7 4	3 1	0	
0 0 0 0	0x0	0	0 0 0 0	0x0	0	0 0 0 0	0x0	0	0 0 0 0	0x0	0	Reset

GPIO_EXT_ETM_TASK_GPIO*n*_EN (*n*: 12 - 15) Configures whether or not to enable GPIO*n* to response ETM task.

0: Not enable

1: Enable

(R/W)

GPIO_EXT_ETM_TASK_GPIO*n*_SEL (*n*: 12 - 15) Configures to select an ETM task channel for GPIO*n*.

0: Select channel 0

1: Select channel 1

......

7: Select channel 7

(R/W)

Register 9.53. GPIO_EXT_ETM_TASK_P4_CFG_REG (0x00B0)

(reserved)	GPIO_EXT_ETM_TASK_GPIO19_SEL	GPIO_EXT_ETM_TASK_GPIO19_EN	(reserved)	GPIO_EXT_ETM_TASK_GPIO18_SEL	GPIO_EXT_ETM_TASK_GPIO18_EN	(reserved)	GPIO_EXT_ETM_TASK_GPIO17_SEL	GPIO_EXT_ETM_TASK_GPIO17_EN	(reserved)	GPIO_EXT_ETM_TASK_GPIO16_SEL	GPIO_EXT_ETM_TASK_GPIO16_EN	
31 28	27 25	24	23 20	19 17	16	15 12	11 9	8	7 4	3 1	0	
0 0 0 0	0x0	0	0 0 0 0	0x0	0	0 0 0 0	0x0	0	0 0 0 0	0x0	0	Reset

GPIO_EXT_ETM_TASK_GPIO*n*_EN (*n*: 16 - 19) Configures whether or not to enable GPIO*n* to response ETM task.
0: Not enable
1: Enable
(R/W)

GPIO_EXT_ETM_TASK_GPIO*n*_SEL (*n*: 16 - 19) Configures to select an ETM task channel for GPIO*n*.
0: Select channel 0
1: Select channel 1
......
7: Select channel 7
(R/W)

Register 9.54. GPIO_EXT_ETM_TASK_P5_CFG_REG (0x00B4)

(reserved)	GPIO_EXT_ETM_TASK_GPIO23_SEL	GPIO_EXT_ETM_TASK_GPIO23_EN	(reserved)	GPIO_EXT_ETM_TASK_GPIO22_SEL	GPIO_EXT_ETM_TASK_GPIO22_EN	(reserved)	GPIO_EXT_ETM_TASK_GPIO21_SEL	GPIO_EXT_ETM_TASK_GPIO21_EN	(reserved)	GPIO_EXT_ETM_TASK_GPIO20_SEL	GPIO_EXT_ETM_TASK_GPIO20_EN	
31 28	27 25	24	23 20	19 17	16	15 12	11 9	8	7 4	3 1	0	
0 0 0 0	0x0	0	0 0 0 0	0x0	0	0 0 0 0	0x0	0	0 0 0 0	0x0	0	Reset

GPIO_EXT_ETM_TASK_GPIO*n*_EN (*n*: 20 - 23) Configures whether or not to enable GPIO*n* to response ETM task.

0: Not enable

1: Enable

(R/W)

GPIO_EXT_ETM_TASK_GPIO*n*_SEL (*n*: 20 - 23) Configures to select an ETM task channel for GPIO*n*.

0: Select channel 0

1: Select channel 1

......

7: Select channel 7

(R/W)

Register 9.55. GPIO_EXT_ETM_TASK_P6_CFG_REG (0x00B8)

(reserved)	GPIO_EXT_ETM_TASK_GPIO27_SEL	GPIO_EXT_ETM_TASK_GPIO27_EN	(reserved)	GPIO_EXT_ETM_TASK_GPIO26_SEL	GPIO_EXT_ETM_TASK_GPIO26_EN	(reserved)	GPIO_EXT_ETM_TASK_GPIO25_SEL	GPIO_EXT_ETM_TASK_GPIO25_EN	(reserved)	GPIO_EXT_ETM_TASK_GPIO24_SEL	GPIO_EXT_ETM_TASK_GPIO24_EN	
31 28	27 25	24	23 20	19 17	16	15 12	11 9	8	7 4	3 1	0	
0 0 0 0	0x0	0	0 0 0 0	0x0	0	0 0 0 0	0x0	0	0 0 0 0	0x0	0	Reset

GPIO_EXT_ETM_TASK_GPIO*n*_EN (*n*: 24 - 27) Configures whether or not to enable GPIO*n* to response ETM task.

0: Not enable

1: Enable

(R/W)

GPIO_EXT_ETM_TASK_GPIO*n*_SEL (*n*: 24 - 27) Configures to select an ETM task channel for GPIO*n*.

0: Select channel 0

1: Select channel 1

......

7: Select channel 7

(R/W)

Register 9.56. GPIO_EXT_ETM_TASK_P7_CFG_REG (0x00BC)

(reserved)	GPIO_EXT_ETM_TASK_GPIO31_SEL	GPIO_EXT_ETM_TASK_GPIO31_EN	(reserved)	GPIO_EXT_ETM_TASK_GPIO30_SEL	GPIO_EXT_ETM_TASK_GPIO30_EN	(reserved)	GPIO_EXT_ETM_TASK_GPIO29_SEL	GPIO_EXT_ETM_TASK_GPIO29_EN	(reserved)	GPIO_EXT_ETM_TASK_GPIO28_SEL	GPIO_EXT_ETM_TASK_GPIO28_EN	
31 28	27 25	24	23 20	19 17	16	15 12	11 9	8	7 4	3 1	0	
0 0 0 0	0x0	0	0 0 0 0	0x0	0	0 0 0 0	0x0	0	0 0 0 0	0x0	0	Reset

GPIO_EXT_ETM_TASK_GPIO*n*_EN (*n*: 28 - 31) Configures whether or not to enable GPIO*n* to response ETM task.

0: Not enable

1: Enable

(R/W)

GPIO_EXT_ETM_TASK_GPIO*n*_SEL (*n*: 28 - 31) Configures to select an ETM task channel for GPIO*n*.

0: Select channel 0

1: Select channel 1

......

7: Select channel 7

(R/W)

Register 9.57. GPIO_EXT_ETM_TASK_P8_CFG_REG (0x00C0)

Field	(reserved)	GPIO_EXT_ETM_TASK_GPIO35_SEL	GPIO_EXT_ETM_TASK_GPIO35_EN	(reserved)	GPIO_EXT_ETM_TASK_GPIO34_SEL	GPIO_EXT_ETM_TASK_GPIO34_EN	(reserved)	GPIO_EXT_ETM_TASK_GPIO33_SEL	GPIO_EXT_ETM_TASK_GPIO33_EN	(reserved)	GPIO_EXT_ETM_TASK_GPIO32_SEL	GPIO_EXT_ETM_TASK_GPIO32_EN
Bits	31 28	27 25	24	23 20	19 17	16	15 12	11 9	8	7 4	3 1	0
Reset	0 0 0 0	0x0	0	0 0 0 0	0x0	0	0 0 0 0	0x0	0	0 0 0 0	0x0	0

GPIO_EXT_ETM_TASK_GPIO*n*_EN (*n*: 32 - 35) Configures whether or not to enable GPIO*n* to response ETM task.

0: Not enable

1: Enable

(R/W)

GPIO_EXT_ETM_TASK_GPIO*n*_SEL (*n*: 32 - 35) Configures to select an ETM task channel for GPIO*n*.

0: Select channel 0

1: Select channel 1

......

7: Select channel 7

(R/W)

Register 9.58. GPIO_EXT_ETM_TASK_P9_CFG_REG (0x00C4)

(reserved)	GPIO_EXT_ETM_TASK_GPIO39_SEL	GPIO_EXT_ETM_TASK_GPIO39_EN	(reserved)	GPIO_EXT_ETM_TASK_GPIO38_SEL	GPIO_EXT_ETM_TASK_GPIO38_EN	(reserved)	GPIO_EXT_ETM_TASK_GPIO37_SEL	GPIO_EXT_ETM_TASK_GPIO37_EN	(reserved)	GPIO_EXT_ETM_TASK_GPIO36_SEL	GPIO_EXT_ETM_TASK_GPIO36_EN	
31 28	27 25	24	23 20	19 17	16	15 12	11 9	8	7 4	3 1	0	
0 0 0 0	0x0	0	0 0 0 0	0x0	0	0 0 0 0	0x0	0	0 0 0 0	0x0	0	Reset

GPIO_EXT_ETM_TASK_GPIO*n*_EN (*n*: 36 - 39) Configures whether or not to enable GPIO*n* to response ETM task.

0: Not enable

1: Enable

(R/W)

GPIO_EXT_ETM_TASK_GPIO*n*_SEL (*n*: 36 - 39) Configures to select an ETM task channel for GPIO*n*.

0: Select channel 0

1: Select channel 1

......

7: Select channel 7

(R/W)

Register 9.59. GPIO_EXT_ETM_TASK_P10_CFG_REG (0x00C8)

(reserved)	GPIO_EXT_ETM_TASK_GPIO43_SEL	GPIO_EXT_ETM_TASK_GPIO43_EN	(reserved)	GPIO_EXT_ETM_TASK_GPIO42_SEL	GPIO_EXT_ETM_TASK_GPIO42_EN	(reserved)	GPIO_EXT_ETM_TASK_GPIO41_SEL	GPIO_EXT_ETM_TASK_GPIO41_EN	(reserved)	GPIO_EXT_ETM_TASK_GPIO40_SEL	GPIO_EXT_ETM_TASK_GPIO40_EN	
31 28	27 25	24	23 20	19 17	16	15 12	11 9	8	7 4	3 1	0	
0 0 0 0	0x0	0	0 0 0 0	0x0	0	0 0 0 0	0x0	0	0 0 0 0	0x0	0	Reset

GPIO_EXT_ETM_TASK_GPIO*n*_EN (*n*: 40 - 43) Configures whether or not to enable GPIO*n* to response ETM task.

0: Not enable

1: Enable

(R/W)

GPIO_EXT_ETM_TASK_GPIO*n*_SEL (*n*: 40 - 43) Configures to select an ETM task channel for GPIO*n*.

0: Select channel 0

1: Select channel 1

......

7: Select channel 7

(R/W)

Register 9.60. GPIO_EXT_ETM_TASK_P11_CFG_REG (0x00CC)

	(reserved)	GPIO_EXT_ETM_TASK_GPIO47_SEL	GPIO_EXT_ETM_TASK_GPIO47_EN	(reserved)	GPIO_EXT_ETM_TASK_GPIO46_SEL	GPIO_EXT_ETM_TASK_GPIO46_EN	(reserved)	GPIO_EXT_ETM_TASK_GPIO45_SEL	GPIO_EXT_ETM_TASK_GPIO45_EN	(reserved)	GPIO_EXT_ETM_TASK_GPIO44_SEL	GPIO_EXT_ETM_TASK_GPIO44_EN
Bits	31 28	27 25	24	23 20	19 17	16	15 12	11 9	8	7 4	3 1	0
Reset	0 0 0 0	0x0	0	0 0 0 0	0x0	0	0 0 0 0	0x0	0	0 0 0 0	0x0	0

GPIO_EXT_ETM_TASK_GPIO*n*_EN (*n*: 44 - 47) Configures whether or not to enable GPIO*n* to response ETM task.

0: Not enable

1: Enable

(R/W)

GPIO_EXT_ETM_TASK_GPIO*n*_SEL (*n*: 44 - 47) Configures to select an ETM task channel for GPIO*n*.

0: Select channel 0

1: Select channel 1

......

7: Select channel 7

(R/W)

Register 9.61. GPIO_EXT_ETM_TASK_P12_CFG_REG (0x00D0)

(reserved)	GPIO_EXT_ETM_TASK_GPIO51_SEL	GPIO_EXT_ETM_TASK_GPIO51_EN	(reserved)	GPIO_EXT_ETM_TASK_GPIO50_SEL	GPIO_EXT_ETM_TASK_GPIO50_EN	(reserved)	GPIO_EXT_ETM_TASK_GPIO49_SEL	GPIO_EXT_ETM_TASK_GPIO49_EN	(reserved)	GPIO_EXT_ETM_TASK_GPIO48_SEL	GPIO_EXT_ETM_TASK_GPIO48_EN	
31 28	27 25	24	23 20	19 17	16	15 12	11 9	8	7 4	3 1	0	
0 0 0 0	0x0	0	0 0 0 0	0x0	0	0 0 0 0	0x0	0	0 0 0 0	0x0	0	Reset

GPIO_EXT_ETM_TASK_GPIO*n*_EN (*n*: 48 - 51) Configures whether or not to enable GPIO*n* to response ETM task.

0: Not enable

1: Enable

(R/W)

GPIO_EXT_ETM_TASK_GPIO*n*_SEL (*n*: 48 - 51) Configures to select an ETM task channel for GPIO*n*.

0: Select channel 0

1: Select channel 1

......

7: Select channel 7

(R/W)

Register 9.62. GPIO_EXT_ETM_TASK_P13_CFG_REG (0x00D4)

(reserved)	GPIO_EXT_ETM_TASK_GPIO54_SEL	GPIO_EXT_ETM_TASK_GPIO54_EN	(reserved)	GPIO_EXT_ETM_TASK_GPIO53_SEL	GPIO_EXT_ETM_TASK_GPIO53_EN	(reserved)	GPIO_EXT_ETM_TASK_GPIO52_SEL	GPIO_EXT_ETM_TASK_GPIO52_EN	
31 … 20	19 … 17	16	15 … 12	11 … 9	8	7 … 4	3 … 1	0	
0 0 0 0 0 0 0 0 0 0 0 0	0x0	0	0 0 0 0	0x0	0	0 0 0 0	0x0	0	Reset

GPIO_EXT_ETM_TASK_GPIO*n*_EN (*n*: 52 - 54) Configures whether or not to enable GPIO*n* to response ETM task.

0: Not enable

1: Enable

(R/W)

GPIO_EXT_ETM_TASK_GPIO*n*_SEL (*n*: 52 - 54) Configures to select an ETM task channel for GPIO*n*.

0: Select channel 0

1: Select channel 1

......

7: Select channel 7

(R/W)

Register 9.63. GPIO_EXT_VERSION_REG (0x00FC)

(reserved)	GPIO_EXT_DATE	
31 … 28	27 … 0	
0 0 0 0	0x2203050	Reset

GPIO_EXT_DATE Version control register.

(R/W)

9.20.4 LP GPIO Matrix Registers

The addresses in this section are relative to LP GPIO matrix base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

For how to program reserved fields, please refer to Section IX .

Register 9.64. LP_GPIO_CLK_EN_REG (0x0000)

(reserved)	LP_GPIO_CLK_EN	
31 1	0	
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	1	Reset

LP_GPIO_CLK_EN Configures whether or not to enable clock gate.

0: Not enable

1: Enable, the clock is free running.

(R/W)

Register 9.65. LP_GPIO_VER_DATE_REG (0x0004)

(reserved)	LP_GPIO_VER_DATE	
31 28	27 0	
0 0 0 0	0x230323	Reset

LP_GPIO_VER_DATE Version control register.

(R/W)

Register 9.66. LP_GPIO_OUT_REG (0x0008)

(reserved)	LP_GPIO_OUT_DATA	
31 16	15 0	
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	0	Reset

LP_GPIO_OUT_DATA Configures the output value of GPIO0 ~ GPIO15 output in simple GPIO output mode.

0: Low level

1: High level

The value of bit0 ~ bit15 correspond to the output value of GPIO0 ~ GPIO15 respectively. Bit16 ~ bit31 are invalid.

(R/W/WTC)

Register 9.67. LP_GPIO_OUT_W1TS_REG (0x000C)

(reserved)	LP_GPIO_OUT_DATA_W1TS	
31 ... 16	15 ... 0	
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	0	Reset

LP_GPIO_OUT_DATA_W1TS Configures whether or not to set the output register LP_GPIO_OUT_REG of GPIO0 ~ GPIO15.

0: Not set

1: The corresponding bit in LP_GPIO_OUT_REG will be set to 1

Bit0 ~ bit15 are corresponding to GPIO0 ~ GPIO15. Bit16 ~ bit31 are invalid. Recommended operation: use this register to set LP_GPIO_OUT_REG.

(WT)

Register 9.68. LP_GPIO_OUT_W1TC_REG (0x0010)

(reserved)	LP_GPIO_OUT_DATA_W1TC	
31 ... 16	15 ... 0	
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	0	Reset

LP_GPIO_OUT_DATA_W1TC Configures whether or not to clear the output register LP_GPIO_OUT_REG of GPIO0 ~ GPIO15 output.

0: Not clear

1: The corresponding bit in LP_GPIO_OUT_REG will be cleared.

Bit0 ~ bit15 are corresponding to GPIO0 ~ GPIO15. Bit16 ~ bit31 are invalid. Recommended operation: use this register to clear LP_GPIO_OUT_REG.

(WT)

Register 9.69. LP_GPIO_ENABLE_REG (0x0014)

(reserved)	LP_GPIO_ENABLE_DATA	
31 … 16	15 … 0	
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	0	Reset

LP_GPIO_ENABLE_DATA Configures whether or not to enable the output of GPIO0 ~ GPIO15.

0: Not enable

1: Enable

Bit0 ~ bit15 are corresponding to GPIO0 ~ GPIO15. Bit16 ~ bit31 are invalid.

(R/W/WTC)

Register 9.70. LP_GPIO_ENABLE_W1TS_REG (0x0018)

(reserved)	LP_GPIO_ENABLE_DATA_W1TS	
31 … 16	15 … 0	
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	0	Reset

LP_GPIO_ENABLE_DATA_W1TS Configures whether or not to set the output enable register LP_GPIO_ENABLE_REG of GPIO0 ~ GPIO15.

0: Not set

1: The corresponding bit in LP_GPIO_ENABLE_REG will be set to 1

Bit0 ~ bit15 are corresponding to GPIO0 ~ GPIO15. Bit16 ~ bit31 are invalid. Recommended operation: use this register to set LP_GPIO_ENABLE_REG.

(WT)

Register 9.71. LP_GPIO_ENABLE_W1TC_REG (0x001C)

(reserved)	LP_GPIO_ENABLE_DATA_W1TC	
31 … 16	15 … 0	
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	0	Reset

LP_GPIO_ENABLE_DATA_W1TC Configures whether or not to clear the output enable register LP_GPIO_ENABLE_REG of GPIO0 ~ GPIO15.

0: Not clear

1: The corresponding bit in LP_GPIO_ENABLE_REG will be cleared

Bit0 ~ bit15 are corresponding to GPIO0 ~ 15. Bit16 ~ bit31 are invalid. Recommended operation: use this register to clear LP_GPIO_ENABLE_REG.

(WT)

Register 9.72. LP_GPIO_STATUS_REG (0x0020)

(reserved)	LP_GPIO_STATUS_DATA	
31 … 16	15 … 0	
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	0	Reset

LP_GPIO_STATUS_DATA The interrupt status of GPIO0 ~ GPIO15, can be configured by the software.

Bit0 ~ bit15 are corresponding to GPIO0 ~ GPIO15. Bit16 ~ bit31 are invalid.

Each bit represents the status of its corresponding GPIO:

0: Represents the GPIO does not generate the interrupt configured by LP_GPIO_PIN*n*_INT_TYPE, or this bit is configured to 0 by the software.

1: Represents the GPIO generates the interrupt configured by LP_GPIO_PIN*n*_INT_TYPE, or this bit is configured to 1 by the software.

(R/W/WTC)

Register 9.73. LP_GPIO_STATUS_W1TS_REG (0x0024)

	(reserved)	LP_GPIO_STATUS_DATA_W1TS	
Bits	31 … 16	15 … 0	
	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	0	Reset

LP_GPIO_STATUS_DATA_W1TS Configures whether or not to set the interrupt status register LP_GPIO_STATUS_DATA of GPIO0 ~ GPIO15.

Bit0 ~ bit15 are corresponding to GPIO0 ~ GPIO15. Bit16 ~ bit31 are invalid.

If the value 1 is written to a bit here, the corresponding bit in LP_GPIO_STATUS_DATA will be set to 1.

Recommended operation: use this register to set LP_GPIO_STATUS_DATA.

(WT)

Register 9.74. LP_GPIO_STATUS_W1TC_REG (0x0028)

	(reserved)	LP_GPIO_STATUS_DATA_W1TC	
Bits	31 … 16	15 … 0	
	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	0	Reset

LP_GPIO_STATUS_DATA_W1TC Configures whether or not to clear the interrupt status register LP_GPIO_STATUS_DATA of GPIO0 ~ GPIO15.

Bit0 ~ bit15 are corresponding to GPIO0 ~ GPIO15. Bit16 ~ bit31 are invalid.

If the value 1 is written to a bit here, the corresponding bit in LP_GPIO_STATUS_DATA will be cleared.

Recommended operation: use this register to clear LP_GPIO_STATUS_DATA.

(WT)

Register 9.75. LP_GPIO_STATUS_NEXT_REG (0x002C)

(reserved)	LP_GPIO_STATUS_INTERRUPT_NEXT	
31 … 16	15 … 0	
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	0x00	Reset

LP_GPIO_STATUS_INTERRUPT_NEXT Represents the interrupt source signal of GPIO0 ~ GPIO15.
Bit0 ~ bit15 are corresponding to GPIO0 ~ 15. Bit16 ~ bit31 are invalid. Each bit represents:
0: The GPIO does not generate the interrupt configured by LP_GPIO_PIN*n*_INT_TYPE.
1: The GPIO generates an interrupt configured by LP_GPIO_PIN*n*_INT_TYPE.
The interrupt could be rising edge interrupt, falling edge interrupt, level sensitive interrupt and any edge interrupt.
(RO)

Register 9.76. LP_GPIO_IN_REG (0x0030)

(reserved)	LP_GPIO_IN_DATA_NEXT	
31 … 16	15 … 0	
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	0	Reset

LP_GPIO_IN_DATA_NEXT Represents the input value of GPIO0 ~ GPIO15. Each bit represents a pin input value:
0: Low level
1: High level
Bit0 ~ bit15 are corresponding to GPIO0 ~ GPIO15. Bit16 ~ bit31 are invalid.
(RO)

Register 9.77. LP_GPIO_PIN*n*_REG (*n*: 0 - 15) (0x0034+4**n*)

(reserved)	LP_GPIO_PIN*n*_EDGE_WAKEUP_CLR	LP_GPIO_PIN*n*_PAD_DRIVER	LP_GPIO_PIN*n*_INT_TYPE	LP_GPIO_PIN*n*_WAKEUP_ENABLE	
31 … 6	5	4	3 … 1	0	
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	0	0	0	0	Reset

LP_GPIO_PIN*n*_WAKEUP_ENABLE Configures whether or not to enable GPIO wake-up function.

0: Disable

1: Enable

This function only wakes up system from Deep-sleep.

(R/W)

LP_GPIO_PIN*n*_INT_TYPE Configures GPIO interrupt type.

0: GPIO interrupt disabled

1: Rising edge trigger

2: Falling edge trigger

3: Any edge trigger

4: Low level trigger

5: High level trigger

(R/W)

LP_GPIO_PIN*n*_PAD_DRIVER Configures to select pin drive mode.

0: Normal output

1: Open drain output

(R/W)

LP_GPIO_PIN*n*_EDGE_WAKEUP_CLR Configures to clear edge-triggered GPIO wake-up source in PMU.

1: The corresponding bit in LP_PMU_xxx will be cleared.

0: No effect.

(WT)

Register 9.78. LP_GPIO_FUNC*n*_IN_SEL_CFG_REG (*n*: 0 - 15) (0x0074+4n*)**

(reserved)	LP_GPIO_FUNC*n*_IN_SEL	LP_GPIO_SIG*n*_IN_SEL	LP_GPIO_FUNC*n*_IN_INV_SEL	
31 … 8	7 … 2	1	0	
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	0x30	0	0	Reset

LP_GPIO_FUNC*n*_IN_INV_SEL Configures whether or not to invert the input value.

0: Not invert

1: Invert

(R/W)

LP_GPIO_SIG*n*_IN_SEL Configures whether or not to route signals via LP GPIO matrix.

0: Bypass LP GPIO matrix, i.e., connect signals directly to peripheral configured in LP IO MUX.

1: Route signals via LP GPIO matrix.

(R/W)

LP_GPIO_FUNC*n*_IN_SEL Configures to select a pin from the 16 LP GPIO pins to connect the input signal *n*.

0: Select GPIO0

1: Select GPIO1

......

14: Select GPIO14

15: Select GPIO15

16 ~ 31: Invalid

32 ~ 47: A constantly low input

48 ~ 63: A constantly high input

(R/W)

Register 9.79. LP_GPIO_FUNC*n*_OUT_SEL_CFG_REG (*n*: 0 - 15) (0x00F4+4n*)**

(reserved)	LP_GPIO_FUNCn_OUT_SEL	LP_GPIO_FUNCn_OUT_INV_SEL	LP_GPIO_FUNCn_OE_SEL	LP_GPIO_FUNCn_OE_INV_SEL	
31 ... 9	8 ... 3	2	1	0	
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	0x20	0	0	0	Reset

LP_GPIO_FUNC*n*_OE_INV_SEL Configures whether or not to invert the output enable signal.

0: Not invert

1: Invert

(R/W)

LP_GPIO_FUNC*n*_OE_SEL Configures to select the source of output enable signal.

0: Use output enable signal from peripheral.

1: Force the output enable signal to be sourced from bit*n* of LP_GPIO_ENABLE_REG.

(R/W)

LP_GPIO_FUNC*n*_OUT_INV_SEL Configures whether or not to invert the output value.

0: Not invert

1: Invert

(R/W)

LP_GPIO_FUNC*n*_OUT_SEL Configures to select a signal *Y* (0 <= *Y* < 32) from 31 peripheral signals to be output to GPIO*n*.

0: Select signal 0

1: Select signal 1

......

30: Select signal 30

31: Select signal 31

Or

32: Bit*n* of LP_GPIO_OUT_REG and LP_GPIO_ENABLE_REG are selected as the output value and output enable.

For the detailed signal list, see Table 9.13-1.

(R/W)

9.20.5 LP IO MUX Registers

The addresses in this section are relative to LP IO MUX base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

For how to program reserved fields, please refer to Section IX .

Register 9.80. LP_IOMUX_CLK_EN_REG (0x0000)

	(reserved)	LP_IOMUX_CLK_EN
Bits	31 … 1	0
Reset	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	1

LP_IOMUX_CLK_EN Configures whether or not to enable clock gate.

0: Not enable

1: Enable, the clock is free running.

(R/W)

Register 9.81. LP_IOMUX_VER_DATE_REG (0x0004)

	(reserved)	LP_IOMUX_VER_DATE
Bits	31 … 28	27 … 0
Reset	0 0 0 0	0x230313

LP_IOMUX_VER_DATE Version control register.

(R/W)

Register 9.82. LP_IOMUX_PADn_REG (n = 0 - 15) (0x0008+4*n)

Bits	31–12	11	10	9	8	7	6–5	4	3	2	1–0
Field	(reserved)	LP_IOMUX_PADn_FILTER_EN	LP_IOMUX_PADn_FUN_IE	LP_IOMUX_PADn_SLP_OE	LP_IOMUX_PADn_SLP_IE	LP_IOMUX_PADn_SLP_SEL	LP_IOMUX_PADn_FUN_SEL	LP_IOMUX_PADn_MUX_SEL	LP_IOMUX_PADn_RUE	LP_IOMUX_PADn_RDE	LP_IOMUX_PADn_DRV
Reset	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	0	0	0	0	0	0	0	0	0	2

LP_IOMUX_PADn_DRV Configures the drive strength of GPIOn.
0: ~5 mA
1: ~10 mA
2: ~20 mA
3: ~40 mA
(R/W)

LP_IOMUX_PADn_RDE Configures whether or not enable pull-down resistor of GPIOn.
0: Disable
1: Enable
(R/W)

LP_IOMUX_PADn_RUE Configures whether or not to enable pull-up resistor of GPIOn.
0: Disable
1: Enable
(R/W)

LP_IOMUX_PADn_MUX_SEL Configures to decide GPIOn is used by HP IO MUX or LP IO MUX.
0: GPIOn is used by HP IO MUX
1: GPIOn is used by LP IO MUX
(R/W)

LP_IOMUX_PADn_FUN_SEL Configures to select LP IO MUX function for this signal.
0: Select Function 0
1: Select Function 1
......
(R/W)

LP_IOMUX_PADn_SLP_SEL Configures whether or not to enter sleep mode for GPIOn.
0: Not enter
1: Enter
(R/W)

LP_IOMUX_PADn_SLP_IE Configures whether or not to enable the input of GPIOn during sleep mode.
0: Disable
1: Enable
(R/W)

Continued on the next page...

Register 9.83. LP_IOMUX_PAD*n*_REG (*n* = 0 - 15) (0x0008+4*n)

Continued from the previous page...

LP_IOMUX_PAD*n*_SLP_OE Configures whether or not to enable the output of GPIO*n* in sleep mode.

0: Disable

1: Enable

(R/W)

LP_IOMUX_PAD*n*_FUN_IE Configures whether or not to enable input of GPIO*n*.

0: Disable

1: Enable

(R/W)

LP_IOMUX_PAD*n*_FILTER_EN Configures whether or not to enable filter for pin input signals.

0: Disable

1: Enable

(R/W)

Register 9.84. LP_IOMUX_LP_PAD_HOLD_REG (0x004C)

(reserved)	LP_IOMUX_LP_GPIO_HOLD	
31 ... 16	15 ... 0	
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	0	Reset

LP_IOMUX_LP_GPIO_HOLD Configures whether or not to enable hold function of GPIO*n*.

0: Disable

1: Enable

(R/W)

Register 9.85. LP_IOMUX_LP_PAD_HYS_REG (0x0050)

(reserved)	LP_IOMUX_LP_GPIO_HYS	
31 … 16	15 … 0	
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	0	Reset

LP_IOMUX_LP_GPIO_HYS Configures whether or not to enable hysteresis function of GPIO*n*.

0: Disable

1: Enable

Bit0 bit15 are corresponding to GPIO0 GPIO15.

(R/W)